

RESEARCH ARTICLE SUMMARY

NEUROSCIENCE

A thalamocortical circuit encoding deviations from sensory history

Yi Ning Leow, Arundhati Natesan, Alexandria Barlowe, Sofie Åhrlund-Richter, Tianyu (Cindy) Luo, Mehrdad Jazayeri, Mriganka Sur*



Full article and list of author affiliations:
<https://doi.org/10.1126/science.aeg4720>

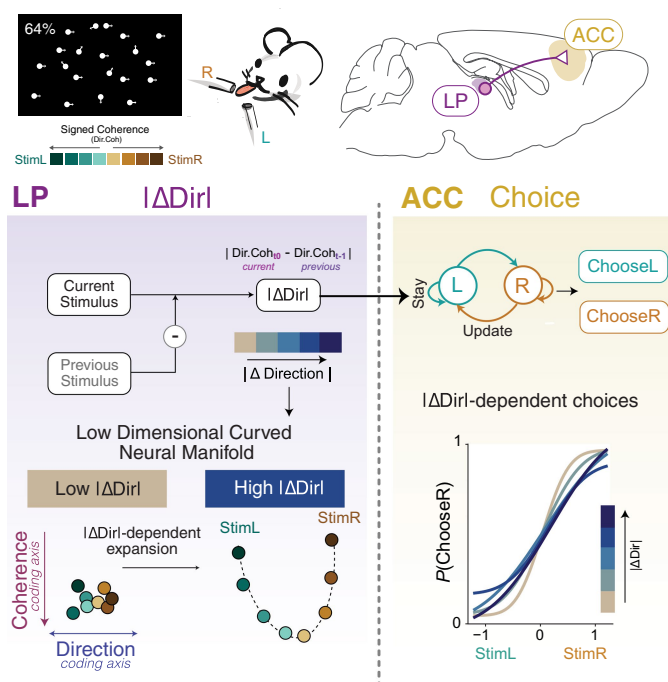
INTRODUCTION: Prior experience can powerfully shape ongoing perception and behavior. But how recent experience alters the brain's processing of ongoing sensory input remains unclear. The higher-order thalamic lateral posterior nucleus (LP; rodent pulvinar) is known to contribute to selective visual processing, whereas the anterior cingulate cortex (ACC) is implicated in evaluating evidence and guiding action, which makes the LP-ACC pathway a compelling candidate for linking recent experience to ongoing decisions.

RATIONALE: We trained mice on a random dot motion discrimination task, which allowed us to quantify the trial-to-trial difference between current and previous motion evidence ($|\Delta\text{Dir}|$) and test how ongoing stimulus input is evaluated relative to recent sensory history. Optogenetic stimulation was performed to address whether perturbing LP-ACC input was sufficient to bias how evidence is converted into choice as a function of $|\Delta\text{Dir}|$. We then used two-photon imaging of LP-ACC axons to examine the structure of the information conveyed by this pathway, investigating whether recent sensory history reshapes the population representation of current motion evidence. These approaches enabled us to test specific computational roles for the LP-ACC pathway in comparing current and recent sensory evidence and to formulate a theoretical framework linking this computation to how current evidence is weighted and when choices are updated.

RESULTS: Mice performed the task with their choices varying with trial-to-trial differences in motion evidence ($|\Delta\text{Dir}|$), indicating graded comparison across successive stimuli rather than a simple serial choice bias. Optogenetic activation of LP-ACC axons unilaterally biased choices toward the stimulated side in a manner that scaled with $|\Delta\text{Dir}|$, which shows that this pathway influences graded history-dependent evaluation of current sensory evidence rather than inducing a motor choice bias.

Two-photon imaging revealed that LP-ACC axons transform current visual evidence into a task-gated low-dimensional population representation, with stimulus evidence organized along a curved trajectory. This representation jointly encoded stimulus direction and coherence and expanded with greater $|\Delta\text{Dir}|$, increasing the separation between stimuli when current evidence deviated more strongly from recent history. This provided evidence that LP-ACC axons encode current stimuli relative to recent sensory history. LP-ACC axons implemented a contrastive code across successive stimuli, such that peak responses were evoked when the current stimulus carried strong evidence for their preferred direction and was maximally different from the previous stimulus. Our imaging results further established that unlike ACC neurons, LP-ACC axons encoded information in a stimulus- rather than choice-centered reference frame.

To understand how stimulus-anchored LP signals could lead to optogenetically induced choice biases, we developed a model with simulated perturbations to explain how LP-derived $|\Delta\text{Dir}|$ signals to



The LP-ACC encodes current sensory evidence relative to recent history. In a two-alternative choice random dot direction discrimination task, the LP provides the ACC with inputs that represent the difference in evidence between current and previous stimuli ($|\Delta\text{Dir}|$), emphasizing deviations through history-dependent expansion of a curved neural manifold representing current evidence. Greater $|\Delta\text{Dir}|$ expands manifold representations and promotes updating. These inputs influence choice arbitration at the ACC, resulting in $|\Delta\text{Dir}|$ -dependent psychometric behavior. StimL, left stimulus; StimR, right stimulus; $P(\text{ChooseR})$, probability of choosing right.

the ACC could influence the weighting of current evidence, probability of choice updating, and direction of choice.

CONCLUSION: Our findings identify the LP-ACC as a thalamocortical pathway that implements a history-dependent contrastive computation, emphasizing deviations from recent sensory history to shape decision updating. More broadly, our results highlight how the higher-order thalamus can contribute to cognitive computation by embedding recent experience into ongoing sensory representations to guide behavior. □

*Corresponding author. Email: msur@mit.edu Cite this article as Y. N. Leow et al., *Science* 393, eaeg4720 (2026). DOI: 10.1126/science.aeg4720

NEUROSCIENCE

A thalamocortical circuit encoding deviations from sensory history

Yi Ning Leow^{1,2,3}, Arundhati Natesan¹, Alexandria Barlowe¹,
Sofie Åhrlund-Richter¹, Tianyu (Cindy) Luo²,
Mehrdad Jazayeri^{2,4}, Mriganka Sur^{1,2*}

Prior expectations guide attention and support perceptual filtering during decision-making. In mice performing a visual discrimination task, choices depended on trial-by-trial differences between consecutive stimuli ($|\Delta\text{Dir}|$). We hypothesized that thalamic lateral posterior (LP; rodent pulvinar) projections to prefrontal areas such as the anterior cingulate cortex (ACC), previously implicated in selective attention and predictive processing, could support history-dependent evaluation of current sensory evidence. In this work, we report that optogenetic manipulations of LP-ACC axons disrupted history-dependent evaluation of current sensory evidence, producing $|\Delta\text{Dir}|$ -dependent choice biases. Two-photon imaging showed that LP-ACC axons represented stimuli along a task-dependent low-dimensional curved manifold whose geometry scaled with $|\Delta\text{Dir}|$, emphasizing larger deviations from recent evidence. These findings identify the LP-ACC as a thalamocortical pathway that implements a contrastive, history-referenced representation of sensory evidence during decisions.

Our perceptual landscape is constantly shaped by prior expectations and ongoing goals, which exert powerful influence over attentional and perceptual processing (1–4). Such expectation-dependent modulation can bias how incoming sensory evidence is evaluated, prioritizing information that is relevant given recent experience or current goals. Beyond transient attentional gain, such influences may operate at the level of decision formation, shaping how sensory information is used to guide choice. The pulvinar, a higher-order visual thalamic nucleus, is well positioned to support such computations through widespread reciprocal connectivity with visual, associative, and frontal cortices. Mutual interactions between the pulvinar and the frontal-parietal attention network have been implicated in regulating sustained and selective attentional allocation during visual decision-making (5–22). Prior work has established compelling evidence for pulvinar contributions to visuospatial attention and distractor filtering, particularly through interactions with the visual cortex and parietal cortex (11–22). However, it remains unclear how pulvinar outputs to the frontal cortex contribute to decision-related processes beyond established roles in attention.

The lateral posterior nucleus (LP) is the rodent homolog of the primate pulvinar, and it shares most of its key cortical partners (23, 24), which indicates that the pulvinar's hub-like functions are conserved across species (8, 12, 15). Recent work has suggested that LP outputs to different cortical areas could carry distinct visual and contextual information (25), underscoring a need for further target-specific functional dissection of information carried by LP outputs. In this work, we focus on LP projections to the anterior cingulate cortex (ACC), a prefrontal region critical for action selection, error computations, and anticipating cues that provide information and resolve uncertainty

(26–34). Reciprocal LP-ACC circuits in rodents integrate inputs from comparable cortical and subcortical regions as the primate pulvinar-frontal circuits (21). Moreover, relative to LP inputs to other visual areas, LP-ACC pathways preferentially integrate inputs from the prefrontal and association areas (23, 24), positioning this pathway to incorporate internal, history, or reward-related signals alongside incoming sensory information. This raises the key question of how LP-ACC interactions influence the evaluation of sensory evidence during decision-making.

We developed a behavioral and circuit-level approach to examine how both prior evidence and current evidence are integrated in LP or pulvinar projections to the ACC. Using a visual discrimination task in mice, we investigated how trial-to-trial differences in sensory evidence shape choice behavior and examined how LP-ACC inputs contribute causally to the process. Using optogenetic perturbations during task performance and two-photon imaging of LP axons in the ACC, we investigated how history-dependent signals are represented in this pathway and how its disruption alters decision-making. Together, our results identify a thalamocortical circuit that links trial history to the evaluation of ongoing sensory evidence, providing a contrastive signal that governs history-dependent evidence evaluation.

Results

Mice make choices on the basis of estimated differences across consecutive stimuli

We trained mice on a two-choice visual discrimination task to test how trial-to-trial differences in sensory evidence influence perceptual decisions. Head-fixed mice learned to discriminate between two global dot movement directions (0° temporally or 180° nasally) in random dot kinematograms (RDKs) presented for 1.5 s and reported their choice by licking either the left or right lick spout during a 1.5-s response window (Fig. 1, A and B). Stimulus uncertainty was varied by presenting RDKs at graded values of coherence (100, 64, 32, or 16% coherence) with equal probabilities. The task was self-paced, requiring mice to withhold licking and run on a treadmill for 1 s to initiate trials. Correct responses were rewarded with water, whereas errors were punished with a time-out.

Trained mice showed coherence-dependent psychometric performance (Fig. 1C). Although choices were largely dependent on the current stimulus evidence, mice nevertheless showed serial choice biases, with a propensity to repeat previously rewarded choices (Fig. 1D). We found that the probability of repeating a choice was highly dependent on the absolute difference between signed coherences across trials, which we named $|\Delta\text{Dir}|$ (Fig. 1, E and F, and fig. S1, A to C). When the current and previous stimuli were more similar (low $|\Delta\text{Dir}|$), mice were more likely to repeat their previously rewarded choice. Conversely, when consecutive stimuli were most different (high $|\Delta\text{Dir}|$), they were least likely to repeat. The opposite was observed after error trials, revealing an outcome-dependent use of $|\Delta\text{Dir}|$ (fig. S1, E and F). This $|\Delta\text{Dir}|$ -dependent strategy extends beyond classical win-stay-lose-switch strategies by explicitly contrasting current and previous evidence and cannot be explained by static choice biases, performance at different stimulus difficulties, or stimulus presentation order (fig. S1, K and L).

Trial-to-trial sensory differences strongly modulated psychometric performance (Fig. 1G). After correct trials, increasing $|\Delta\text{Dir}|$ was associated with reduced hit rates (Fig. 1K), shallower psychometric slopes (Fig. 1L), and increased lapse rates (Fig. 1M), with little effect on decision threshold (fig. S1M). These effects were symmetric with respect to previous choice (Fig. 1H and fig. S2, A to D). Performance varied nonmonotonically with $|\Delta\text{Dir}|$ (Fig. 1, K and L), peaking when consecutive stimuli were similar and declining at intermediate values, consistent with a switch-related cost that is only offset when sensory differences become large. After errors, we observed the opposite trend from after rewarded trials—hit rates improved and lapses decreased with greater $|\Delta\text{Dir}|$ (fig. S1, G to J).

Trial-by-trial choice behavior was best fitted by models incorporating $|\Delta\text{Dir}|$ interactions with evidence and previous choice, outperforming

¹The Picower Institute for Learning and Memory (PILM), Massachusetts Institute of Technology (MIT), Cambridge, MA, USA. ²Department of Brain and Cognitive Sciences (BCS), MIT, Cambridge, MA, USA. ³Institute of Molecular and Cell Biology (IMCB), Agency for Science, Technology and Research (A*STAR), Singapore. ⁴McGovern Institute of Brain Research (MIBR), MIT, Cambridge, MA, USA. *Corresponding author. Email: msur@mit.edu

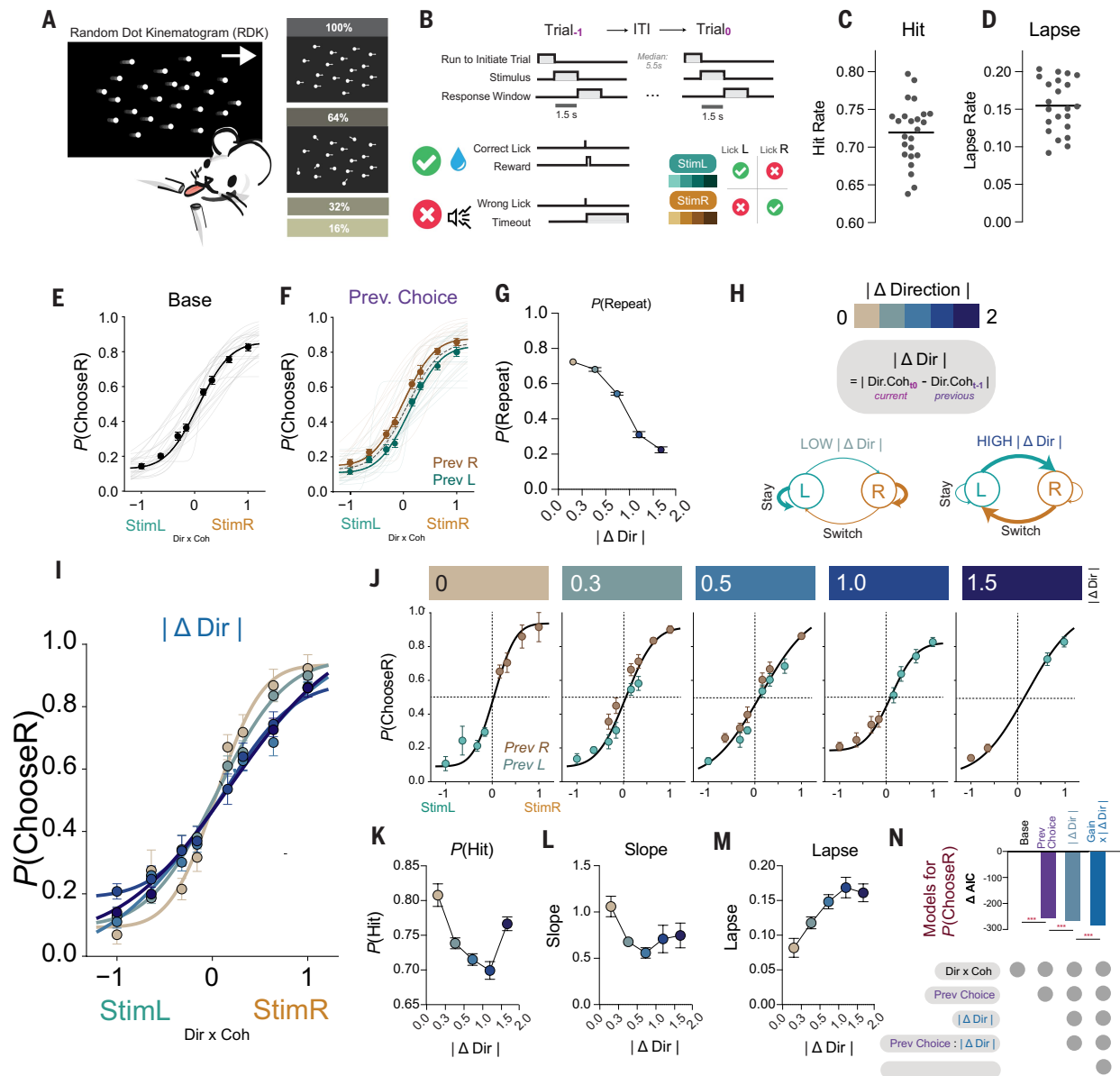


Fig. 1. Mice make choices on the basis of estimated differences across consecutive stimuli. (A) Schematic of the two-alternative choice RDK visual discrimination task. Head-fixed mice reported perceived motion direction by licking a left or right spout. Motion direction (0° temporal or 180° nasal) and coherence (16 to 100%) varied across trials. (B) Trial structure and epochs. Mice run to initiate a trial and are presented with an RDK stimulus for 1.5 s. Mice report their choices by licking either the left or right spout during the 1.5-s response window. Then, 100 ms after choice, correct choices are rewarded with water delivery, whereas wrong choices are followed by a loud white noise and flashing screen. Successive trials are separated by a median ITI of 5.5 s. (C) Hit rates, sampling uniformly from all coherence values. (D) Mean lapses. For (C) and (D), each dot represents a mouse, and the bold line indicates population mean across $N = 24$ mice. (E) Probability of choosing right [$P(\text{ChooseR})$] as a function of signed stimulus strength (direction $\times$ coherence). Population mean is shown in bold black. Gray lines indicate individual animals. $N = 24$ mice. (F) Psychometric functions conditioned on previous choice indicating choice history biases. (G) Probability of repeating the previous choice [$P(\text{Repeat})$] as a function of the absolute change in stimulus direction between consecutive trials ($|\Delta\text{Dir}|$). (H) Schematic illustrating the definition of $|\Delta\text{Dir}|$ as the absolute difference in signed stimulus direction between the current and previous trial and its relationship to stay versus switch behavior. (I) Population-level psychometric functions stratified by $|\Delta\text{Dir}|$. $N = 22$ mice. (J) Psychometric functions separated by previous choice (PrevL versus PrevR) within each $|\Delta\text{Dir}|$ bin. (K to M) Hit rate [$P(\text{Hit})$] (K), psychometric slopes (L), and lapse rate (M), as a function of $|\Delta\text{Dir}|$. (N) Best fit model [lowest Akaike information criterion (AIC)] for $P(\text{ChooseR})$ incorporated $|\Delta\text{Dir}|$ interactions with previous choice terms and evidence. Full details of models are provided in table S3. All error bars indicate SEMs across mice.

models that simply accounted for the previous choice (Fig. 1N and table S3). Thus, mice performed best when consecutive stimuli were similar and when repeating their previously rewarded choices, and they performed worse when they were required to switch. Increasing $|\Delta\text{Dir}|$ also increased the probability of mice omitting the trial (fig. S1N) and was also associated with greater response latency (fig. S1O).

Together, these results implied that mice retained a memory of the previous stimulus estimate, and the comparison with the current stimulus facilitated decisions to switch or stay, on the basis of the previous outcome. Such short-term memory could be expected to decay with time, and indeed $|\Delta\text{Dir}|$ had a diminishing influence on the probability of choice repetition with increasing intertrial intervals (ITIs) (fig. S1, P to S).

This $|\Delta\text{Dir}|$ -dependent strategy was evident across all stimulus coherences rather than selectively enhanced at low coherence. Although modulation of choice repetition was strongest at high coherence (fig. S2E), its effect on performance was comparable across coherence values (fig. S2F), indicating no selective low-coherence performance benefit. Taken together, our findings demonstrate that mice use relative evidence across consecutive trials to modulate classical win-stay-lose-switch strategies.

Activating LP-ACC projections biases choice in a $|\Delta\text{Dir}|$ -dependent manner

The comparison of sensory evidence across trials requires a circuit that integrates both current sensory input and recent evidence, to ultimately influence action selection. To test whether LP inputs to the ACC causally contribute to this process, we optogenetically stimulated LP axons in the ACC. LP-ACC axons expressing the light-activated opsin ChR2 unilaterally were activated by illuminating the ACC with blue light during visual stimulus presentation (Fig. 2, A and B, and fig. S3A). Visual stimuli were always presented on the right screen, which enabled us to dissociate effects driven by contralateral visual input from effects on evidence-to-choice transformation.

Unilateral stimulation of left LP-ACC axons led to an overall impairment in hit rates (Fig. 2, C and D) without a corresponding shift in decision threshold (Fig. 2E), arguing against disruption by simple motor biases. Stimulation of right LP-ACC axons—ipsilateral to stimulus presentation—did not affect performance [$P = 0.0593$, binomial generalized linear mixed-effects model (GLMM); Fig. 2, G and H]. As with left stimulation, decision thresholds remained unchanged (Fig. 2I). Optogenetic activation of either hemisphere significantly reduced psychometric slopes (Fig. 2J; $P = 0.0223$ two-tailed paired t test), indicating a selective reduction in sensitivity and influence of sensory evidence on choice formation rather than a fixed bias. By contrast, both lapse rates (Fig. 2K) and rate of trial omissions (fig. S3B) were unchanged, which suggests that stimulation did not induce attentional disengagement or motor errors. We next asked whether the reduction in psychometric slope after LP-ACC perturbation reflected history-dependent modulation of evidence sensitivity. When we stratified trials by $|\Delta\text{Dir}|$, optogenetic-induced choice biases scaled with increasing contrast between current and previous sensory evidence. Specifically, during left LP-ACC stimulation, mice became less likely to make rightward choices [reduced $P(\text{ChooseR})$] as $|\Delta\text{Dir}|$ increased (Fig. 2, L and M, and fig. S3, F to H). Conversely, right LP-ACC axonal stimulation increased the likelihood of rightward choices with greater $|\Delta\text{Dir}|$ (Fig. 2N and fig. S3, I to K). When pooled across stimulation hemispheres, unilateral optogenetic stimulation of LP-ACC axons increasingly biased choices toward action associated with the side of optogenetic stimulation (ipsilateral) as $|\Delta\text{Dir}|$ increased (Fig. 2O). This $|\Delta\text{Dir}|$ -dependent scaling suggested that stimulation interacts with trial-to-trial sensory differences rather than producing a global motor bias.

Consistent with this, models incorporating a $|\Delta\text{Dir}|$ -dependent, hemisphere-specific bias ($|\Delta\text{Dir}|:\text{OptoSide}$; model f) provided a better account of the data compared with models based on fixed side bias (model b) or previous choice modulation (model e; Fig. 2P and table S4). However, optogenetic stimulation also produced a global reduction in psychometric slope (Fig. 2J), which raises the possibility that stimulation alters sensitivity to current sensory evidence in addition to generating a $|\Delta\text{Dir}|$ -dependent bias. To test whether these effects reflect separable mechanisms, we compared models, including a global evidence interaction ($\text{DirCoh}:\text{Opto}$), a $|\Delta\text{Dir}|$ -dependent lateralized bias ($|\Delta\text{Dir}|:\text{OptoSide}$), or both terms simultaneously. Each interaction independently improved model fit, and the best-fitting model required both $\text{DirCoh}:\text{Opto}$ and $|\Delta\text{Dir}|:\text{OptoSide}$ terms (model h; table S4). Thus, LP-ACC stimulation operates through two separable components—a $|\Delta\text{Dir}|$ -dependent, hemisphere-specific bias and a change in evidence weighting. The direction of this effect depended on the stimulated

hemisphere and scaled with $|\Delta\text{Dir}|$, revealing a lateralized mechanism by which sensory history governs decision updating.

Whereas unilateral LP-ACC perturbation produced detrimental lateralized effects that scaled with the difference between successive stimuli, bilateral stimulation abolished this lateralization (fig. S4). This contrast is consistent with unilateral optogenetic effects arising from asymmetric disruption of a normally bilateral LP-ACC input.

Disruption of $|\Delta\text{Dir}|$ -dependent decision-making was specific to perturbing LP-ACC inputs. Illumination at 593 nm, which does not activate ChR2, produced no behavioral effects (fig. S5), which argues against nonspecific light artifacts. Optogenetically stimulating left LP axons in the primary visual cortex (V1) also did not affect behavior or reproduce $|\Delta\text{Dir}|$ -dependent choice biases (fig. S6), suggesting that the observed effects did not reflect indirect activation of other cortices or nonspecific disruption of visual processing.

Behavioral performance recovered after optogenetic trials regardless of trial outcome (fig. S7), and optogenetic activation during the reinforcement epoch had no effect on subsequent choices (fig. S8), indicating that transient effects were confined to evidence evaluation, without disrupting memory of prior stimuli or outcomes. Taken together, these optogenetic results show that LP-ACC activity biases decisions in a $|\Delta\text{Dir}|$ -dependent manner by selectively shaping how current sensory evidence is evaluated on the basis of recent sensory context.

LP-ACC axons represent visual stimuli with task-gated low-dimensional geometry

The optogenetic results suggest that LP-ACC inputs causally influence how sensory evidence guides choice in light of recent sensory history, rather than providing a simple feedforward visual signal. To determine the nature of information provided by the LP, we performed two-photon imaging of LP-ACC axons during task performance and passive viewing (Fig. 3A and fig. S9). During task engagement, LP-ACC axons exhibited robust, stimulus-evoked responses (Fig. 3B and fig. S11, A and B) that collectively discriminated stimuli defined by direction and coherence, whereas this stimulus information was much weaker under passive viewing in naïve mice (Fig. 3C). This task dependence suggests that LP-ACC activity reflects behaviorally relevant sensory variables rather than bottom-up visual responses alone.

Whereas decoding demonstrates that LP-ACC activity distinguishes stimulus direction, the heterogeneous tuning for direction and coherence across individual LP axons (Fig. 3B and fig. S11) raised the question of how graded evidence for stimulus could be represented across the LP-ACC population. To address this, we summarized trial-averaged population activity for each stimulus using cross-validated principal components analysis (PCA) [first three principal components (PCs): 34.7% total explained variance]. During task performance, LP-ACC population responses varied smoothly and in an orderly manner with graded changes in stimulus direction and strength, giving rise to a curved trajectory aligned with signed stimulus coherence (Fig. 3D). We refer to this continuous curved trajectory as a low-dimensional curved manifold. Along this manifold, population activity jointly reflected stimulus direction and stimulus difficulty rather than clustering into discrete stimulus categories (fig. S10A). This low-dimensional geometry in LP-ACC activity was not evident in naïve mice during passive viewing despite identical sensory input (Fig. 3E). Moreover, the same structure was observed on omission trials lacking overt licking behavior (fig. S10), which rules out a movement-related origin.

As a point of comparison to established cortical decision signals, we examined population activity in ACC neurons during the same task. Consistent with prior work, ACC populations exhibited low-dimensional structure during the task (Fig. 3, F and G, and fig. S10, F, G, J, and K), but the timing and alignment of this structure differed markedly from LP-ACC inputs. During stimulus evaluation, LP-ACC axons exhibited a pronounced expansion of population geometry, reflected by larger spread and longer arc length (Fig. 3, H and I). Increased spread indicates

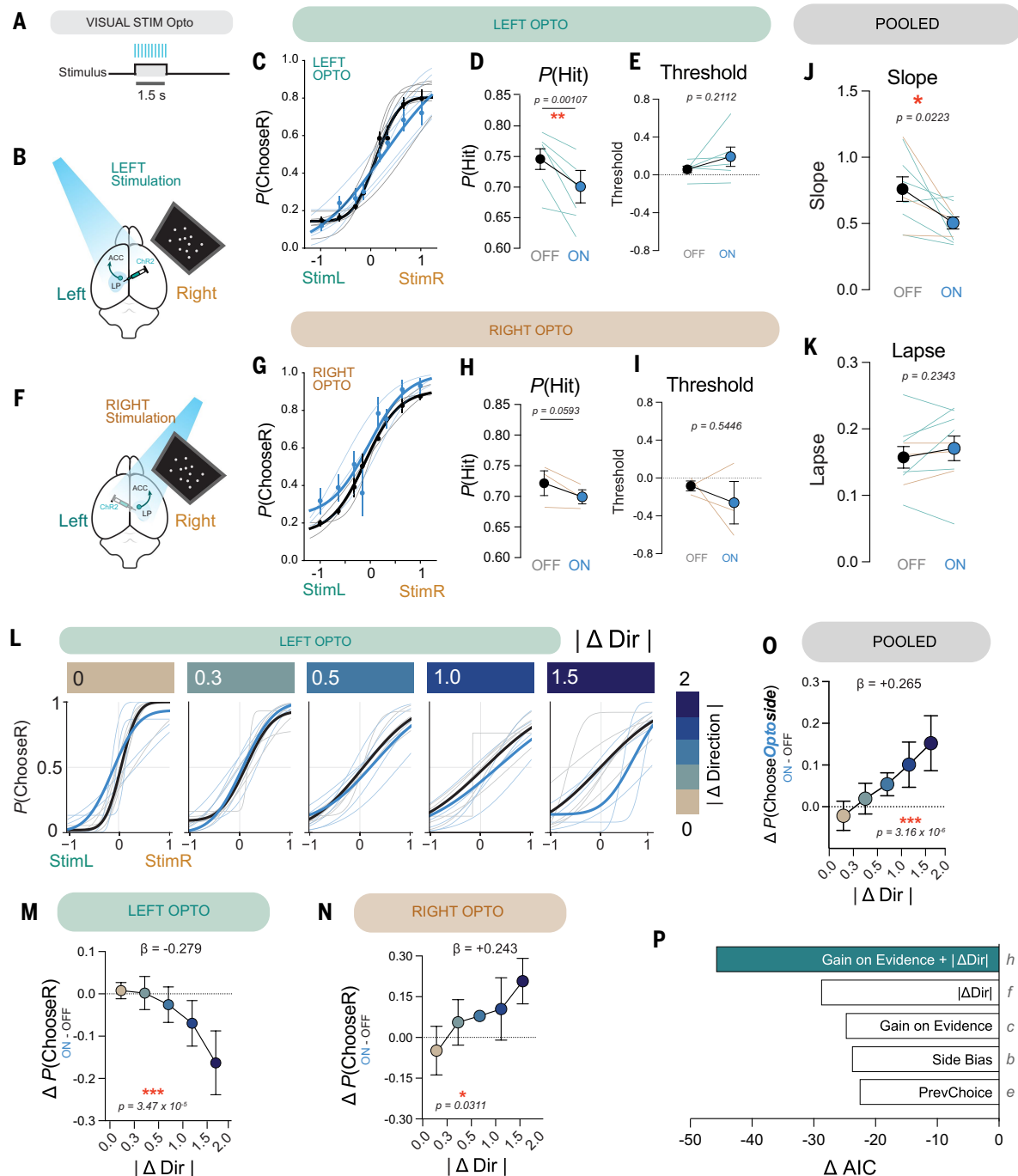


Fig. 2. Activating LP-ACC projections biases choice in a $|\Delta \text{Dir}|$ -dependent manner. (A) Trial timeline showing visual stimulus presentation (1.5 s) and optogenetic stimulation delivered during the stimulus period in a random 25 to 30% of trials. (B) Schematic of left LP axons expressing ChR2 stimulated over the ACC. (C) Psychometric functions [P(ChooseR)] for left LP-ACC opto-stimulation trials (ON; blue) and interleaved control trials (OFF; black). Thin lines indicate individual animals, and thick lines indicate population mean. (D and E) Hit rate [P(Hit)] (D) and psychometric threshold (E) upon left LP-ACC stimulation. (F to I) Same as (B) to (E) but for optogenetic stimulation of right LP axons over the ACC. (J and K) Psychometric slope (J) and lapse rate (K) upon unilateral LP-ACC stimulation, pooled across stimulation sides. (L) Psychometric functions for left LP-ACC stimulation trials, stratified by $|\Delta \text{Dir}|$ (absolute change in stimulus direction from the previous trial). (M and N) $\Delta P(\text{ChooseR})$ as a function of $|\Delta \text{Dir}|$ for left (M) and right (N) LP-ACC stimulation. (O) Change in probability of choosing the optogenetically stimulated side [$\Delta P(\text{ChooseOptoSide})$] as a function of $|\Delta \text{Dir}|$, pooled across stimulation sides. All error bars indicate SEMs across mice; [(D) to (J) and (M)] $n = 6$; [(H) to (K) and (N)] $n = 3$; (O) $n = 9$. Discrete $|\Delta \text{Dir}|$ bins are used only for visualization purposes, and statistical analyses for ChooseR used trial-level data with continuous $|\Delta \text{Dir}|$ values (table S1). (P) Model comparison by ΔAIC relative to a baseline model without any Opto term (table S4, model a). Models are ordered by best fit on top. Models incorporating optogenetic modulation of evidence gain (model c) or $|\Delta \text{Dir}|$ -dependent bias (model f) improved fit relative to static side bias (model b) or previous choice-dependent modulation (model e). The best-fitting model (model h) is highlighted and combined global gain modulation of sensory evidence (DirCoh:Opto) with side-specific $|\Delta \text{Dir}|$ bias (AbsDiff:OptoSide). Full likelihood ratio statistics and model comparisons are reported in table S4.

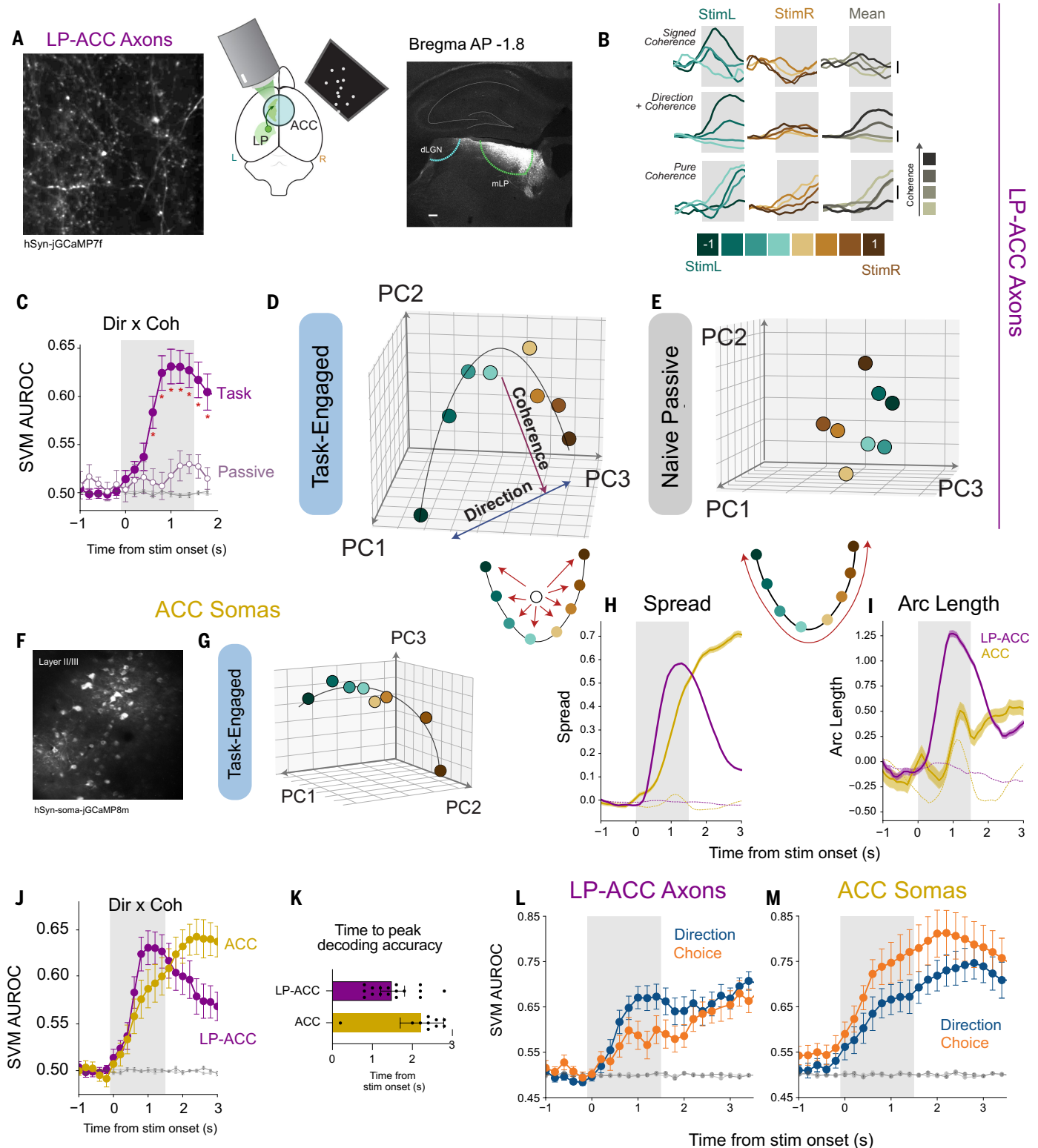


Fig. 3. LP-ACC axons represent visual stimuli along a curved low-dimensional manifold. (A) Schematic of two-photon calcium imaging of LP axons in the ACC during task performance, and example field of view showing GCaMP-expressing LP axons in the ACC (left) and representative confocal micrograph showing medial LP targeting (right). (B) Example trial-averaged traces of single LP-ACC axons with activity that scales with signed coherence (top), with preferential activity or left stimuli, but that also scales monotonically with stimulus coherence (middle) and representing pure coherence encoding without direction discriminability (bottom). (C) Cross-validated SVM decoder performance (AUROC) trained to discriminate each of the eight presented stimuli (two directions $\times$ four coherences) perform much better in the task-engaged condition compared with in naive mice passively viewing identical stimuli. Red asterisks indicate significant difference ($P < 0.05$) in decoding performance at that time point using two-tailed t test. (D) LP-ACC population activity projected onto the first three PCs for distinct presented stimuli when mice were engaged in the task. The representation organizes into a curved manifold that simultaneously represents stimulus direction and coherence. (E) Same as (D) but in naive mice passively viewing identical stimuli; the LP-ACC population does not

have apparent curved manifold organization. **(F)** Schematic of two-photon calcium imaging of the ACC during task performance and example field of view showing GCaMP-expressing LP axons in the ACC. **(G)** ACC population activity projected onto the first three PCs during task engagement. **(H and I)** Manifold metrics with prestimulus baseline subtracted. Spread (H), representing dispersion of stimulus representations from centroid, and arc length (I), which measures path distance through ordered signed coherence, for the LP-ACC (purple) and ACC (yellow). Shading indicates SEMs across iterations. **(J)** Cross-validated SVM decoder performance (AUROC) like in (C) but comparing with ACC (yellow) populations. **(K)** Comparison of time to peak decoding accuracy for $\text{Dir} \times \text{Coh}$ decoders in (J). Each dot represents a single imaging session. **(L and M)** Cross-validated SVM decoder performance (AUROC) decoding binary direction (blue) or choice (orange) for LP-ACC axons (L) and ACC somas (M). [(C), (J), (L), and (M)] Error bars indicate SEMs across sessions. For task-engaged LP-ACC decoding, $n = 17$ sessions, 5 mice [(C) and (L)]; for passive LP-ACC decoding, $n = 13$ sessions, 5 mice (C). For task-engaged ACC decoding, $n = 9$ sessions, 3 mice [(J) and (M)].

greater separation of stimulus representations, whereas longer arc length reflects more extensive and orderly monotonic organization of graded sensory evidence along the curved manifold (Materials and methods).

Spread was consistently larger in LP-ACC populations than in the ACC during the stimulus period, whereas ACC spread increased later to exceed the LP-ACC pathway at its peak (Fig. 3H). By contrast, arc length remained larger in the LP-ACC throughout the stimulus epoch, whereas the ACC exhibited more modest increase later (Fig. 3I).

These temporal and geometric differences were also reflected in decoding analyses. Decoding of graded sensory evidence peaked during stimulus presentation for LP-ACC axons, whereas ACC decoding peaked later near the response (Fig. 3, J and K). Moreover, direction decoding dominated in LP-ACC axons during stimulus evaluation, whereas ACC populations showed stronger choice than direction decoding during the response period (Fig. 3, L and M, and fig. S10, L and M), highlighting a temporal shift from sensory-aligned to choice-aligned population representations across the LP-ACC pathway.

Sensory history reshapes LP-ACC population geometry during stimulus evaluation

Having established that LP-ACC population activity is preferentially aligned with sensory variables during evidence evaluation, we next investigated how recent sensory history modulates representations. A prerequisite for such modulation is that information about the previous stimulus is available during the period when current sensory evidence is evaluated. Consistent with this, previous stimulus direction could be reliably decoded from LP-ACC population activity during the ITT and early in the subsequent trial (Fig. 4A). Previous stimulus direction decoding generalized across trial outcomes—even when the animal's choice was incorrect—indicating dominance of sensory rather than choice history (fig. S12A). Decoding accuracy declined during stimulus presentation, consistent with a transition from history-dominated to stimulus-dominated representations.

The persistence of previous sensory information into the early stimulus period indicates that LP-ACC populations concurrently represent recent and current sensory evidence. Encoding models further revealed widespread multiplexing of current and previous trial variables in single axons that could not be explained by motor correlates alone (fig. S13), motivating a population-level analysis of how task variables are integrated within LP-ACC representations.

Behavioral and optogenetic results identified trial-to-trial differences in sensory evidence ($|\Delta\text{Dir}|$) as a key determinant of choice. We therefore asked whether LP-ACC representations of current sensory evidence are systematically reorganized as a function of $|\Delta\text{Dir}|$ and whether this modulation acts along the same dimensions that encode sensory evidence itself. To address this, we used targeted dimensionality reduction (TDR), a method that identifies the population activity axes that best explain variance related to a specific task variable (Fig. 4B). We can then isolate population coding dimensions (CDs) to quantify how activity along these dimensions varies with $|\Delta\text{Dir}|$. Current stimulus direction explained the largest fraction of LP-ACC population task-related variance, whereas $\text{PrevDirection}_{\text{CD}}$ explained greater total variance than $\text{PrevChoice}_{\text{CD}}$ (Fig. 4C), reinforcing that LP-ACC history representation is sensory dominant rather than choice dominant.

We first examined activity along the CD corresponding to stimulus direction ($\text{Direction}_{\text{CD}}$). Projections onto $\text{Direction}_{\text{CD}}$ revealed that responses to the same visual stimulus were systematically shifted depending on the magnitude of change in stimulus direction relative to the previous trial ($|\Delta\text{Dir}|$; Fig. 4, D to F). These shifts occurred along the same CD used to represent current sensory evidence rather than along a separate history-specific or orthogonal axis.

We next investigated how this axis-level modulation shapes the global geometry of sensory representations. In the joint $\text{Direction}_{\text{CD}}$ and $\text{Coherence}_{\text{CD}}$ subspace, LP-ACC population activity formed a curved manifold with a structure that depended on $|\Delta\text{Dir}|$ (Fig. 4G). When the stimulus direction was similar to the previous (low $|\Delta\text{Dir}|$), representations were more compact, whereas increasing $|\Delta\text{Dir}|$ was associated with progressive expansion of the manifold (Fig. 4H). Consistent with this expansion, the Euclidean distance between left- and right-associated stimulus representations increased monotonically with $|\Delta\text{Dir}|$ (Fig. 4I), indicating enhanced representational separation when current sensory input deviated more strongly from recent history. Beyond category separation, representations also spanned a larger extent of the manifold with increasing $|\Delta\text{Dir}|$, as reflected by increases in trajectory spread and arc length (Fig. 4, J and K).

A comparable history-dependent expansion of representational geometry was not observed in ACC neural populations analyzed in the same manner (fig. S14), which suggests that LP-ACC axons provide a history-referenced input that is transformed, rather than directly preserved, in ACC population activity. Together, these results show that trial-to-trial change in sensory evidence is encoded as a graded reorganization of LP-ACC population geometry, preferentially expanding representational space for larger deviations from recent sensory context.

Contrastive coding of stimulus history in the LP-ACC pathway prioritizes trial-to-trial change

The history-dependent expansion of LP-ACC population geometry raises a key question: What coding structure gives rise to a history-dependent rescaling of sensory representations rather than a uniform shift? To address this, we examined the TDR-defined population coding weights to determine how current and previous sensory evidence are combined across LP-ACC axons.

Population weights defining the $\text{Direction}_{\text{CD}}$ and $\text{PrevDirection}_{\text{CD}}$ axes were systematically anticorrelated (Fig. 5A), indicating opponent contributions of current and recent sensory evidence at the population level. This organization predicts that sensory history will rescale stimulus-evoked responses rather than introduce a uniform offset. To illustrate how this population-level opponency is reflected in axonal responses, we examined population-averaged activity of StimR-selective LP-ACC axons grouped by aligned versus contrastive coding weights to the same stimulus. Axons with aligned weights exhibited similar response polarity for current and previous stimulus direction, consistent with a shift-like influence of history (Fig. 5, B and C). By contrast, axons with contrastive weights showed opposing modulation, such that responses become suppressed when the previous stimulus direction matched the current stimulus (Fig. 5, D and E). The opponent structure was specific to sensory history: Direction and Choice weights were positively aligned (fig. S15A), whereas coupling between Direction and PrevChoice weights was

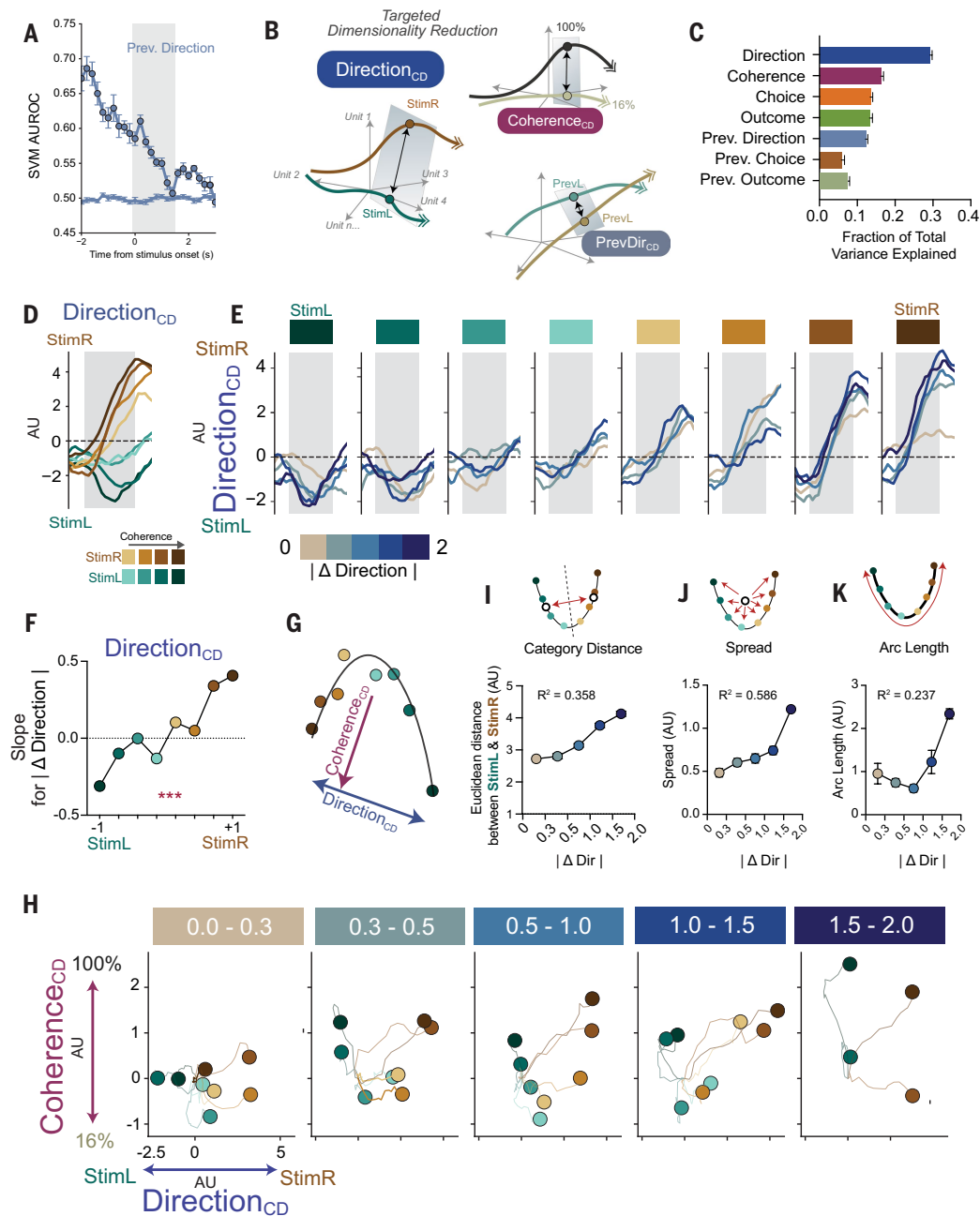


Fig. 4. Sensory history reshapes LP-ACC population geometry during stimulus evaluation. (A) Cross-validated SVM decoder performance (AUROC) decoding of previous direction (blue). Error bars indicate SEMs across $n = 17$ imaging sessions (5 mice). (B) Schematic of TDR analysis used to define task-relevant CDs, such as direction, coherence, and previous direction, from LP-ACC population activity. (C) Fraction of total variance explained by each task CD in LP-ACC axonal populations. (D) Projection of LP-ACC population activity onto the Direction_{CD} shows graded representation of stimulus-signed coherence. AU, arbitrary units. (E) Trial-averaged projections onto the Direction_{CD} for each of the eight distinct stimuli, separated by $|\Delta\text{Dir}|$ (absolute change in stimulus direction from the previous trial), showing systematic shifts with increasing $|\Delta\text{Dir}|$. (F) Quantification of projection shifts along Direction_{CD} as a function of $|\Delta\text{Dir}|$ for left- and right-associated stimuli. Modulation as quantified by slopes for $|\Delta\text{Dir}|$ for each stimulus. Slope was derived from least squares fit for $|\Delta\text{Dir}|$ to mean projection (AU). One-way repeated measures ANOVA on effect of stimulus direction on slopes [$P = 1.65 \times 10^{-43}$, $F(7,525) = 37.9$, $\eta^2 = 0.336$, over 75 iterations]. (G) Schematic illustrating the Direction_{CD} and Coherence_{CD} from TDR that recapitulates the curved manifold observed in PCA. (H) Population trajectories projected into the joint Direction_{CD}-Coherence_{CD} subspace (AU), revealing a curved manifold whose structure expands with $|\Delta\text{Dir}|$. Large dots represent each trajectory at the end of the stimulus evaluation period (1.5 s after stimulus onset), and lines represent their trajectory from baseline (subtracted) upon stimulus onset. (I–K) Manifold metrics. (I) Category distance, measured as Euclidean distance between mean StimL and StimR representations, increases monotonically as a function of $|\Delta\text{Dir}|$. (J) Spread, representing dispersion of stimulus representations from centroid, also increases as a function of $|\Delta\text{Dir}|$. (K) Arc length, path distance through ordered signed coherence, has a U-shaped relationship as a function of $|\Delta\text{Dir}|$. Error bars indicate SEMs across 75 cross-validated TDR iterations.

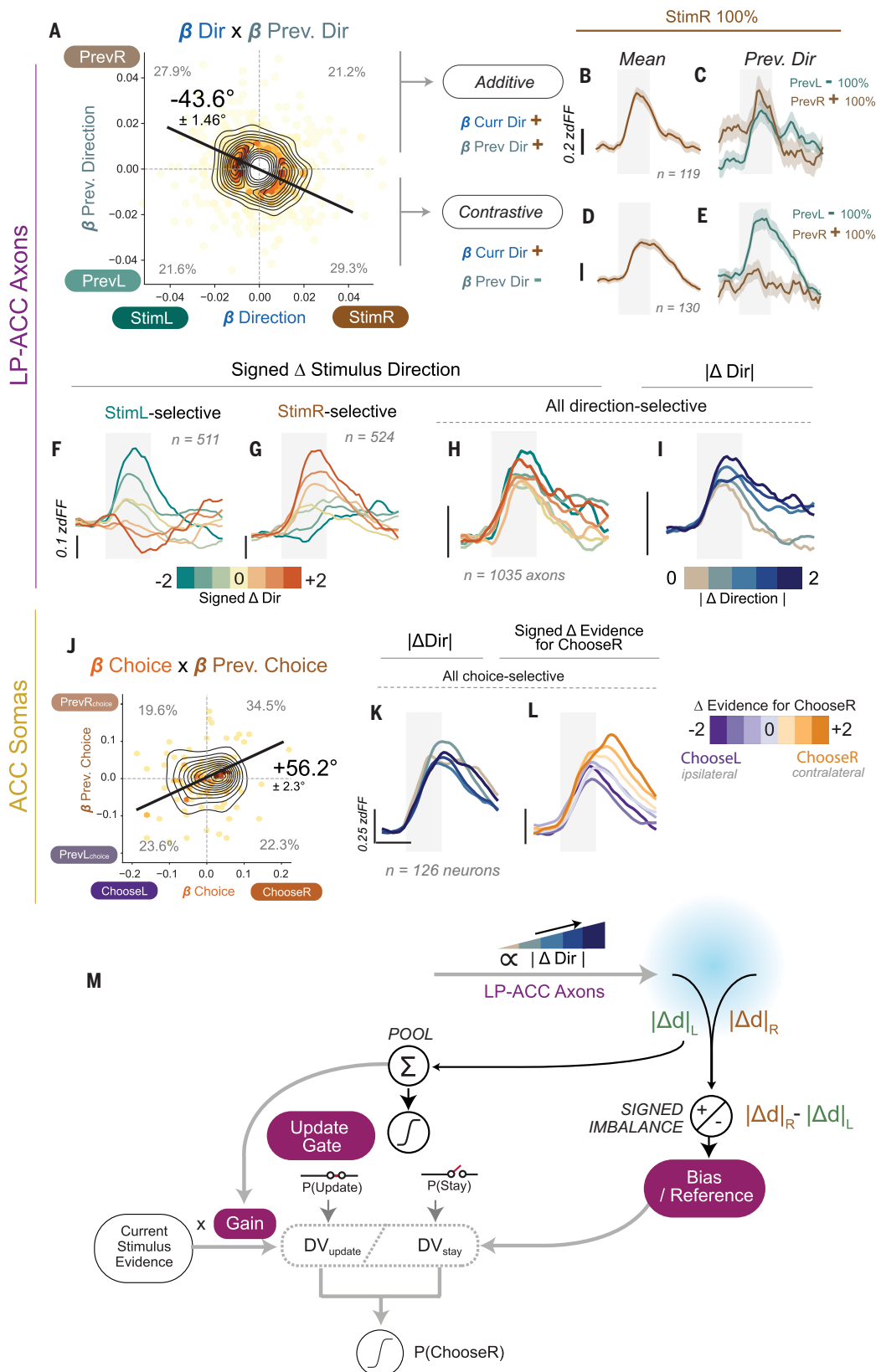


Fig. 5. Contrastive sensory history coding in the LP-ACC supports history-dependent decision updating. (A) Negative correlation between LP-ACC axons' current stimulus direction ($\beta \text{ Dir}$) and previous stimulus direction ($\beta \text{ Prev.Dir}$) coefficients indicate opponent encoding. Only the top 50% of weights are shown for visualization purposes; all analyses used the complete set of weights. Darker colors indicate higher unit density. Percentages indicate the proportion of axons in each quadrant. Major axis angle is the mean $\pm$ SEM across 100 TDR iterations. (B and C) LP-ACC axons with additive aligned weights, selective for both StimR ($\beta \text{ Dir} +$) and PrevR ($\beta \text{ Prev.Dir} +$), responses averaged across all StimR 100% trials (B) or stratified by previous direction at 100% coherence (C). Because weights are aligned and additive, PrevR responses to the same consecutive

stimulus were greater. **(D and E)** Same as **(B)** and **(C)**, but for LP-ACC axons with contrastive weights, which suppressed responses to repeated stimuli while enhancing responses at maximal difference from history. **[(B) to (E)]** Scale bars, 0.2 z-scored dFF. **(F and G)** Average activity of StimL-selective **(F)** and StimR-selective **(G)** LP-ACC axons, plotted by signed ΔDir relative to preferred tuning. **(H)** Pooled direction-selective axons do not show monotonic modulation by signed ΔDir at the level of bulk responses. **(I)** Unsigned $|\Delta\text{Dir}|$ is reflected in bulk responses of pooled direction-selective axons. **[(F) to (I)]** Scale bars, 0.1 z-scored dFF. **(J)** Correlation between ACC neurons coefficients for current (β Choice) and previous choice (β Prev.Choice) show additive coding of choice history. Plot details as in **(A)**. **(K)** Choice-selective ACC neurons do not show monotonic scaling with unsigned $|\Delta\text{Dir}|$. **(L)** Choice-selective axons are modulated by signed $\Delta\text{evidence}$ for contralateral choice (ChooseR). **[(K) and (L)]** Scale bars, 0.25 z-scored dFF. **(M)** Model schematic. LP-ACC axons encode trial-to-trial differences in sensory evidence as signed ΔDir or unsigned $|\Delta\text{Dir}|$, provided independently to the ACC in each hemisphere ($|\text{dL}|$ and $|\text{dR}|$). These signals contribute to decision-making through three computations. First, its pooled magnitude governs entry into an update versus stay regime (“update gate”). Second, they scale sensory evidence gain (“gain”) in the update regime. Third, asymmetric inputs lead to interhemispheric imbalance in the ACC, generating a signed “bias/reference signal.” Simulated optogenetic perturbation alters these upstream signals and accounts for $|\Delta\text{Dir}|$ -dependent switching, unilateral lateralized effects, and bilateral cancellation (supplementary text, note 1).

weak and inconsistent (fig. S15, B and E), arguing against a generic correlation across task variables.

When responses were grouped by stimulus tuning, both StimL- and StimR-selective axons showed systematic scaling with signed stimulus difference relative to their preferred direction, with no systematic bias for one stimulus direction over the other (Fig. 5, F and G). When activity was pooled across axons, signed stimulus-difference information was not prominently expressed in bulk averages (Fig. 5H), despite remaining linearly separable in population space (Fig. 4). By contrast, pooling emphasized the magnitude of trial-to-trial sensory change ($|\Delta\text{Dir}|$; Fig. 5I), which we previously observed to be associated with an expansion of the population manifold and increased separation between stimulus representations (Fig. 4). Together, these analyses show that LP-ACC population activity jointly conveys stimulus identity and sensory change magnitude: Stimulus identity is embedded in the geometry of population responses, whereas $|\Delta\text{Dir}|$ is reflected both through geometric expansion and aggregate population activity.

We next turned to the ACC, where population activity is dominated by choice-related variance (fig. S15H). We therefore examined the relationship between current and previous choice representations. In contrast to the opponent organization in the LP-ACC pathway, ACC neurons showed predominantly aligned, positively correlated coding for current and previous choice (Fig. 5J and fig. S15, I to M), consistent with reinforcement of action history and a shift-like bias in choice representations rather than contrastive integration. Accordingly, unlike the LP-ACC, monotonic scaling with $|\Delta\text{Dir}|$ was no longer apparent in ACC bulk responses (Fig. 5K). Instead, ACC bulk activity scaled monotonically with the signed difference in choice-evidence signal, driven by aligned contributions of current and previous choice as well as the dominance of contralateral (ChooseR) choice-selective neurons (Fig. 5L and fig. S15, N to P). These imaging results delineate a transformation from contrastive, stimulus-centered representations in the LP-ACC pathway to aligned, choice-centered representations in the ACC.

A model linking LP comparison signals to $|\Delta\text{Dir}|$ -dependent choice updating

Integrating the imaging and optogenetic results, we therefore implemented a reduced model in which the LP provides stimulus-anchored comparison signals independently to each hemisphere (Fig. 5M and supplementary text, notes 1 and 2); the information sent to each hemisphere is assumed to be identical. In the model, LP-derived comparison signals play two separable functions: Their pooled magnitude controls entry into an update regime (“Update Gate”), whereas their interhemispheric difference provides a signed imbalance that biases choice direction upon updating (“Bias/Reference”). Because $|\Delta\text{Dir}|$ information sent to each hemisphere is assumed to be identical, the net bias is always zero except when simulating unilateral optogenetic stimulation.

Behavior defaults to a win-stay strategy unless $|\Delta\text{Dir}|$ information provides sufficient drive to enter the update regime—an internal cognitive state, rather than an action state, and thus one that does not necessarily involve choice switching. Within the update regime, the model incorporates increasing $|\Delta\text{Dir}|$ -dependent gain on sensory evidence

(“gain”), consistent with the expansion of stimulus representations observed in LP-ACC population activity. This allows strong and clear differences in evidence to overcome the intrinsic cost of switching and restore performance, and it was a mechanism required to reproduce the nonmonotonic psychometric sensitivity observed in normative behavior (Fig. 1, I to L, and fig. S16, A to F). Thus, behavior emerges from a mixture of persistence (“stay”) and change-responsive (“update”) regimes, in which LP-derived $|\Delta\text{Dir}|$ signals provide information that guides updating without directly encoding choice.

Simulating optogenetic modulation as changes in LP-derived ΔDir comparison signals applied before ACC readout, the model reproduced $|\Delta\text{Dir}|$ -dependent entry into update states, lateralized biases under unilateral perturbation, and cancellation under bilateral stimulation (fig. S16, G to Y). Crucially, the model reproduced the observed $|\Delta\text{Dir}|$ -dependent scaling of unilateral optogenetic effects because larger $|\Delta\text{Dir}|$ increased both update probability and the magnitude of the induced bias through greater interhemispheric mismatch. Under symmetric bilateral perturbation, the pooled comparison magnitude is greater and increases the probability of entering the update regime without any directional bias, such that its net impact on accuracy depends on $|\Delta\text{Dir}|$. Perturbing LP-derived comparison signals before ACC readout was sufficient to account for these effects across various perturbation schemes (fig. S16, G to Y, and supplementary text, note 1). Whereas selective modulation of the bias term alone was sufficient to reproduce the primary lateralized optogenetic phenotype (supplementary text, note 1, and fig. S17B), reproducing the full baseline behavioral structure—including the nonmonotonic psychometric sensitivity (Fig. 1, K and L)—required the mixture of stay and update regimes that governs $|\Delta\text{Dir}|$ -dependent gain (supplementary text, note 1, and fig. S17). Together, these simulations indicate that LP-derived $|\Delta\text{Dir}|$ signals inform when updating occurs and bias its direction, without directly encoding choice.

Discussion

Perceptual decision-making involves comparing current sensory evidence with recent experience, yet how such comparisons are implemented at the circuit level remains to be fully elucidated. In this work, we identify a thalamocortical mechanism in which this comparison is realized through a contrastive population code in LP projections to the ACC. LP-ACC axons encode current sensory evidence relative to the previous trial through an opponent coding structure rather than a uniform shift. As a result, population activity is organized to emphasize changes from recent sensory history: Low-dimensional population manifolds representing stimuli expand with increasing trial-to-trial changes in evidence, embedding information about stimulus identity and change in sensory evidence within population structure. By emphasizing deviations from recent history, this contrastive population code is well-suited to influence when incoming evidence leads to an update.

At a mechanistic level, our results support a role for LP-ACC inputs in reshaping the population representation of sensory evidence through comparison-level gain. Contrastive coding of current relative to recent sensory evidence gives rise to an expansion of population geometry with

increasing differences between successive stimuli, reparameterizing decision-relevant evidence. Critically, this rescaling need not reflect changes in low-level sensory encoding: Rather than globally enhancing or suppressing sensory evidence, LP-ACC signals bias how evidence is represented and compared and may modulate the effective gain of these representations in a history-dependent manner.

Consistent with a comparison-level role, our findings argue against a model in which LP-ACC projections act as a feedforward visual pathway. Unilateral optogenetic perturbation produced opposite-signed $|\Delta\text{Dir}|$ -dependent effects across hemispheres, including when stimulation was applied to the hemisphere lacking direct sensory input (Fig. 2). A strictly feedforward sensory pathway would instead predict effects confined to the sensory-driven hemisphere. This is consistent with the anatomical origin of LP-ACC inputs: Medial pulvinar neurons are enriched for associative, multisensory, and motor-related inputs and are less constrained by visual topography (17, 23, 35) compared with other pulvinar subdivisions embedded within early visual cortical loops (11, 12, 18–22, 36–40).

This circuit-level interpretation also clarifies the temporal locus of LP-ACC influence. History-dependent biases have been described as shifts in initial conditions or bias terms in evidence accumulation models (41) and have been linked to distributed cortical representations of prior information (42). Several cortical regions, most notably the posterior parietal cortex, have been implicated in mediating such biases, with inactivations during the ITI reducing history-dependent influences on subsequent choices (43, 44). By contrast, despite robust representation of previous-trial information in LP-ACC axons during the ITI, ChR2 activation of the LP-ACC pathway during this epoch had little effect on behavior (fig. S8). This suggests that the LP-ACC pathway contributes to history-dependent decision biases through modulation of ongoing evidence evaluation rather than by setting a persistent bias state before stimulus onset.

Consistent with a role during stimulus evaluation, our results indicate that LP-ACC inputs regulate when and how sensory evidence is incorporated to drive updating rather than simply biasing decisions toward or away from recent choices. In both imaging and modeling, contrastive comparison signals scaled with the magnitude of differences in sensory evidence and gated entry into an update regime, linking trial-to-trial comparison to changes in evidence evaluation that ultimately shape action. This comparison-based update away from a default win-stay policy provides a concrete circuit mechanism for history-dependent control of decision updating and is consistent with theoretical frameworks that emphasize predictive and context-dependent modulation of cortical processing by the pulvinar (12, 13, 36–38). We formalized behavior as a mixture of stay and update regimes, with LP-derived $|\Delta\text{Dir}|$ signals particularly important for supporting updating (Fig. 5M and supplementary text, note 1). Nevertheless, whether such history-dependent modulation of evidence leads to behaviorally optimal choices across task contexts or instead reflects circuit-level constraints or priorities independent of task demands remains an open question. Our results instead support a more constrained interpretation in which decisions are shaped by a difference-based evaluation of current sensory evidence relative to recent sensory history. Under this view, LP inputs contribute information relevant for determining when sensory evidence is sufficiently different to support updating a prior choice, but this computation is accompanied by a switch cost. We therefore emphasize that the observed history dependence reflects a mechanistic property of evidence evaluation rather than a flexible strategy designed to optimize behavioral performance.

Both LP-ACC axons and ACC neurons represent sensory evidence in a structured curved geometry (Fig. 3 and fig. S10), extending low-dimensional manifold representations of decision variables beyond cortical circuits (45, 46) to thalamic structures embedded in recurrent decision networks. We observed expansion of geometric representations

that scaled with the difference in sensory evidence across trials in LP-ACC axonal populations (Figs. 4 and 5) but not in the ACC (fig. S14), where history induces more additive, shift-like biases along choice-centered dimensions. This dissociation suggests that a common low-dimensional scaffold for evidence representation can be differentially modulated and transformed in a region-specific manner, reflecting distinct computational roles within the decision network and respective population tuning (47).

LP-ACC input emphasizes contrastive, history-relative sensory evidence, whereas the ACC transforms these inputs into choice-aligned representations through competitive integration. Critically, choice representations in the ACC are lateralized and evaluated through interhemispheric competition, such that balanced activity across hemispheres is required for unbiased choice (48–50). Because LP-ACC inputs are anchored to sensory variables rather than choice, without systematic preference for one side, their influence on behavior must therefore be mediated through how they perturb this competitive ACC readout. Consistent with this framework, we found that unilateral—but not bilateral—perturbation of LP input led to biased choice, mirroring effects of lateralized perturbations previously reported in the ACC. The abolition of bias under bilateral activation indicates that unilateral optogenetic biases depend on interhemispheric imbalance and engage downstream competitive ACC circuitry.

Although the optogenetic results implicate LP-derived comparison signals as a critical, sensory-centered input to ACC decision circuitry, the model does not specify how those signals are altered by stimulation. Rather, behavioral biases emerge through competitive integration as LP-derived signals are transformed within the ACC into action-centered representations. How this transformation is implemented at the microcircuit level remains an open question. LP axons preferentially target superficial ACC layers (24), where they are well positioned to interact with local inhibitory and recurrent circuitry (51–56), providing a plausible substrate through which sensory-centered comparison signals could be converted into action-selective biases.

Viewed in the broader context of pulvinar function, comparison-based rescaling of sensory evidence by LP-ACC inputs aligns with growing evidence that the pulvinar supports predictive and contrastive computations (7, 12, 13, 57). Previous work has shown that pulvinar cortical circuits can compute differences between concurrently competing stimuli (58), encode confidence-related signals (59), or report mismatches between sensory input and self-generated actions such as locomotion or eye movements (22, 36–38). By contrast, our results provide evidence that LP-ACC projections encode how current sensory evidence deviates from internally maintained, history-dependent expectations rather than mismatches between simultaneously present sensory or visuomotor signals.

Prior experience fundamentally shapes how new sensory information is prioritized during decision-making. By encoding current sensory evidence relative to recent history, LP-ACC inputs provide a mechanism for emphasizing deviations from prior experience, thereby shaping when incoming evidence is likely to drive a change in action. Our work positions the pulvinar as an active contributor to perceptual filtering, modulating the influence of sensory evidence on the basis of recent history. Such context-dependent modulation may shape the use of neural processing resources by selectively amplifying changes in sensory input relative to recent history.

Materials and methods

Animals and surgery

All experiments were performed under protocols approved by the MIT Institutional Animal Care and Use Committee (IACUC) and conformed to NIH guidelines. Wild-type C57/BL6J mice were used for all experiments except for ACC imaging under naïve passive viewing, where transgenic jRCaMP8m mice were used (JAX: 037718). Mice were all between 8 and 12 weeks old at time of surgery. Stereotaxic surgeries

were performed under isoflurane anesthesia (4% for induction and 1% for maintenance). Viral injections were performed using glass micropipettes and injected at a rate of 50 to 75 nl/min. 75 nl of AAV1-syn-jGCaMP7f (Addgene:104488-AAV1) or AAV1-hSyn-hChR2(H134R)-EYFP (Addgene: 26973-AAV1) was injected in the left medial LP [relative to bregma: −1.8 mm anteroposterior (AP), −1.2 mm mediolateral (ML), −2.6 mm dorsoventral (DV)] contralateral to the visual stimulus screen. For ACC imaging, 100 nl of AAV1-hSyn-soma-jGCaMP8m (Addgene: 169257-AAV1; for task) or AAV9-CaMKII-Cre (Addgene: 105558-AAV9; diluted 1:1000; for passive) was injected at three to five sites 200 to 400 μ m ventral of the cortical surface of the left ACC, within the cranial window proximal to the sagittal sinus and avoiding blood vessels. 3-mm-wide cranial windows (two 3-mm glass coverslips glued under a 5-mm glass coverslip for stability) were implanted centered on the midline at AP +1 mm (relative to bregma) over ACC/M2. A titanium head-plate was implanted over window to allow head fixation during imaging. Implants were secured with dental cement (CB Metabond) mixed with black ink to render the dental cement opaque. After surgery, mice recovered on a heated water-circulation pad for at least an hour. Mice were injected subcutaneously with slow-release (72 hours) buprenorphine (1 mg/kg) before surgery and meloxicam (5 mg/kg) after surgery for analgesia on the day of surgery. Meloxicam (5 mg/kg) was administered once every 24 hours for 2 days after surgery or longer until mice begin nesting. After surgery, mice were housed in pairs or triplets if there were compatible cage mates, with a small proportion of them requiring single housing. For mice imaged under passive viewing contexts, imaging was performed 3 to 6 weeks after surgery.

Mice trained on the visual discrimination task recovered postsurgery for 7 to 14 days before being placed on water regulation. A running wheel was also introduced into their home cage at this time as enrichment and to habituate them to the running wheel on the rig. Mice were closely monitored to ensure they do not lose more than 10% of their body weight over the span of a week. If so, or if demonstrating signs of poor health, mice were returned to ad lib water for at least a week before a slow taper returning to water regulation with an increased daily water allowance. If mice did not receive all their daily water during a training session, they were supplemented up to their daily water (ranging 1 to 1.5 ml depending on the starting weight of the mouse) with hydrogel.

Visual stimulus presentation

RDK stimuli were generated and presented through PsychoPy on a 22.5 cm $\times$ 14 cm screen at 1360 pixel $\times$ 768 pixel resolution at 60 frames per second. The screen was placed contralateral to the imaged hemisphere, 8 cm away from the right eye of the mouse and oriented $\sim 35^\circ$ relative to the mouse head-body axis. The size of each dot corresponded to 2° visual angle and was always presented at 100% contrast against a black background. Approximately 280 dots were presented in a full field during each trial. Each random dot had a lifetime of 500 ms, after which another dot reappeared randomly in the field moving in the same direction. In presentations where dot motion coherence was varied, the noise dots moved at the same speed in a random (out of eight possible directions) but with constant direction for its lifetime.

In the task, RDKs were presented at uniform speed of 46° /s. In naïve passive conditions, sessions contained presentations with random dots at three different speeds (27° , 46° , or 57° /s). However, for the purpose of comparison with the task context, we only used data from trials with RDKs moving at 46° /s. In the naïve passive condition, each stimulus presentation lasted 1.5 s after a blank interstimulus period of 1.5 s.

In the task condition, the presentation of the visual stimulus was self-paced, with at least a 3-s ITI, a new trial began when the mouse ran at a threshold speed for 1 s and had no detectable licks for at least 1 s. An RDK stimulus was then presented for 1.5 s. Stimuli were presented in pseudorandomly shuffled blocks of eight distinct stimuli (two directions and for each, four coherences: 16, 32, 64, and 100%) to

ensure sessions contained comparable proportions of each distinct stimulus, which is presented at least once every eight trials.

Two-photon imaging and preprocessing

LP-ACC axons were imaged through a 3-mm cranial window over the ACC with an Olympus objective [25 $\times$, 1.05 numerical aperture (NA)] at 4 $\times$ zoom. GCaMP7f was excited at 930-nm wavelength with a Ti sapphire laser. Axons were imaged 100 to 200 μ m below the cortical surface, corresponding to layer I. Images were acquired at 512 $\times$ 381 pixels with resonant scanning at 40 Hz, with frames averaged to a working frame-rate of 10 Hz with which we performed motion correction. Videos were motion-corrected with a correlation-based rigid template registration using ImageJ. Individual axonal bouton regions of interest (ROIs), were manually selected from the maximum Z-projection of the video. A 2-pixel-wide annulus is selected around each ROI as an estimate of the background neuropil signal. The fluorescence signal from the ROI is taken as $F_{ROI} = F_{ROIraw} - 0.7 \times F_{neuropil}$. The signals were smoothed by filtering with a causal Gaussian kernel. The session signals were detrended by fitting a linear regression across the entire time series. For computing dF/F₀ for each ROI, we fit a histogram of the fluorescence signal with a Gaussian kernel and take the fluorescence signal corresponding to the peak of the fitted kernel as the baseline F₀. dF/F₀ was then computed as $(F - F_0)/F_0$. Signals were denoised with the OASIS algorithm (60) as implemented in the CalmAn package (61).

Axonal clustering

To cluster boutons that may potentially be from the same neuronal unit, we performed pairwise correlations between every bouton pair in a session. We then performed hierarchical clustering based on these pairwise temporal correlations. The number of clusters was determined empirically for each session by iteratively testing decreasing distance thresholds to satisfy a cluster correlation threshold where (i) the minimum correlation within a cluster was >0.2 , (ii) at least two times that of the 85th percentile between/intercluster correlation, and (iii) there is no statistical difference between inter- and intracluster correlations when the same distance threshold was applied to a shuffled dataset where individual units were temporally randomly shifted by 500 to 1000 frames (50 to 100 s). We weighted boutons of each cluster by the signal-to-noise ratio of the ROI, taking a weighted average of fluorescence signals to represent bouton clusters (referred to as axons throughout the manuscript).

Optogenetics

Laser stimulation was delivered either using an optical fiber placed over a 3-mm-wide cranial window or through a 400- μ m-wide optic fiber implanted over the ACC. ChR2-expressing axons were stimulated with a 493-nm laser during the visual stimulus period (1.5 s) at 20 Hz (10 mW/mm², 493 nm). For the postreinforcement activation, stimulation was delivered at 10 Hz (10 mW/mm², 493 nm) over the course of 3 s. As within-subject controls, we performed optogenetic stimulation experiments with a 593-nm laser (10 mW/mm²), which fell outside of the range of ChR2 excitation spectrum. For stimulation during the postreinforcement period, we ensured that a new trial could not be initiated until 500 ms after the laser stimulation ended. Laser stimulation was triggered through WaveSurfer (Janelia). Two or three trials were randomly selected in each stimulus block (eight different stimuli) to be laser-on trials (~ 20 to 25% of total trials). For LP-V1 axons, an optic fiber [Thorlabs: 400 μ m fiber, 2.5 mm outside diameter (OD) cannula, 2 mm length, 0.5 NA] was implanted over V1 (bregma AP: −4.0, ML: −3.0), these were intended to allow sufficient light spread over V1 (wider fiber and NA) but localized sufficiently distant from other cortical areas (stereotactic coordinates) that receive LP axons.

Statistics

Statistical analyses were performed using the Python libraries pingouin or statsmodels. Regular and binomial linear mixed-effects models

were fitted in R using the lme4 package. Where there were multiple comparisons, P values were reported after Benjamini-Hochberg's adjustment. Trial-level behavioral outcomes (such as Choice, Hit, Repeat, Omit) were analyzed using GLMMs with a binomial error distribution and logit link function. These models are appropriate for binary outcomes and do not assume normally distributed residuals. All GLMMs included random intercepts for mouse identity to account for repeated measurements across trials within animals. Where the effects of optogenetic stimulation were tested, models for each response variable were constructed to remain minimal but still control for known covariates (such as stimulus coherence for Hits). Full model specifications and predictors are detailed in tables S1 to S4. Group-level analyses of psychometric-derived parameters were performed using linear mixed effects models or repeated-measures analysis of variance (rmANOVA) with $|\Delta\text{Dir}|$ bin as a within-subject factor. Sphericity was assessed using Mauchly's test, and Greenhouse-Geisser corrections were applied when sphericity was violated. Where rmANOVA revealed significant main effects, post hoc comparisons were conducted with appropriate correction for multiple comparisons.

Support vector machine (SVM) decoding

Decoding of stimulus features (direction, coherence, etc.) was performed by training cross-validated linear SVM classifiers on axonal activity in 200-ms time bins. Trials from each session were randomly shuffled, then stratified to have equal proportions of each of the distinct stimuli in training and testing sets, split in a 70:30 ratio. Unless otherwise stated, training sets contained both hit and error trials for task-engaged conditions. Decoding efficacy was evaluated by area under the receiver operator characteristic (AUROC) curve using held out test sets (30% of trials). For omitted trials, only trials where no early (during stimulus) or late (after response period) licking detected were included to exclude motor confounds. We performed 100 iterations for decoding of each session, with random training and testing trial splits in each iteration. Where decoding involves more than two targets, decoders were trained using a one-versus-one strategy to maximally discriminate between all targets, whereas reported AUROC represents the average of all one-versus-rest AUROC scores. For each iteration, decoders were additionally trained and tested on axonal activity with shuffled target labels to serve as controls.

Encoding models

Trials were shuffled and then split threefold into training and testing sets, stratified to have equal proportions of different trial stimuli (eight different types) in training and testing sets. Trials spanning 1 s before and 5 s after stimulus onset were then concatenated to construct the full design matrix. As predictors, we include three temporal factors [ITI (1 s, 10 frames), stimulus (1.5 s, 15 frames), choice (variable number of frames between stimulus offset and reinforcement) and reinforcement (variable number of frames from reinforcement to 5 s after stimulus onset)] that spanned the time bins in each epoch. The motor parameters we included were right lick rates, left lick rates, and running speed, which were included as rolling averages (500-ms window averaging). We included trial-based selectivity for stimulus direction, choice, and trial outcome, each ranging -1 to $+1$ and graded by coherence. To capture the temporal dynamics of each trial predictor, predictors were expanded in time in the form of a Toeplitz matrix. A coefficient was obtained for every time bin of the trial for every predictor, together constituting the fitted response kernel for that predictor. All task selectivity factors were represented from stimulus onset, whereas previous trial parameters were represented from the ITI onward. Each predictor in the design matrix was mean-centered. The DFF of each axon was normalized by its maximum value. Regression fits were regularized using "elasticnet" penalty with a 1:99 L1:L2 regularization ratio. For each axonal unit, we ran 100 iterations of generalized linear model (GLM) fits with different

combinations of training and testing sets, reporting the mean coefficient of determination (R^2) and model coefficients derived from all iterations. We considered a unit task-responsive if the task-dependent model, on held-out test trials, had an R^2 significantly greater than 0.01 across all 100 iterations (one-tailed, one-sample t test, $P < 0.05$). A unit was regarded to encode a specific task variable during that epoch if dropping the task variable from the model caused a significant reduction in explained variance R^2 (one-tailed, one-sample t test, $P < 0.05$) relative to the full model, across 100 iterations.

Dimensionality reduction

For dimensionality reduction by PCA, trials in each session were randomly split, with a half of trials serving as a training set, whereas the other half was held out for cross-validation. The split was stratified by the signed coherence of the stimulus direction to enforce equal proportions of specific stimuli in both fitting and validation sets. Each set was trial-averaged for every distinct stimulus, aligned to the onset. Every unit was mean subtracted and scaled to have unit variance. We then pooled axons from across sessions and mice into a pseudopopulation (N units). PCA was performed on a response matrix composed of pseudopopulation activity in a time window spanning 200 to 1200 ms (T, 10 time bins of 100 ms each) after stimulus onset, concatenating the time windows across each of the eight stimuli (S) ($40 \times N$). The averaged population activity of axons in the held-out cross-validation trials was then projected onto the low-dimensional PC space. Pairwise-Euclidean distances between individual stimulus representations in fig. S10 were averaged from a 1-s window from 200 to 1200 ms from stimulus onset. This window was chosen to begin from when stimulus discriminability was first above chance based on direction decoding results, and to end right before any choice- or action-related activity.

For TDR, we performed cross-validated TDR onto stimulus direction and coherence axes, using a procedure adapted from Mante *et al.* (62). We first mean-centered and scaled the Ca^{2+} data to have zero mean and unit variance. The traces across the entire session were then denoised by applying PCA. Ca^{2+} activity was projected onto the subspace defined by the minimum number of PCs required to retain 75% of the total explained variance, followed by an inverse transformation to reconstruct the denoised data. As for PCA, we performed stratified splits of each session into random halves, with half the trials serving as a fitting set and the other half serving as a validation set. For the DirectionCD, we computed an average difference vector ($N \times 1$) by subtracting the population activity vectors for left versus right stimulus directions, averaged over a time window where we first observed direction discriminability emerge, and before any choice or motor preparatory activity begins (200 to 1200 ms after stimulus onset). For the CoherenceCD, the difference vector was computed between 100% coherence trials and 16% coherence trials. Selectivity vectors were obtained similarly for Choice (Choose Right versus Choose Left), Outcome (Hit versus Error), Previous Direction, Previous Choice, and Previous Outcome, and were organized in this order in the selectivity matrix after Direction and Coherence. The selectivity vectors for each task variable were then orthogonalized using QR decomposition. After orthogonalization, we projected the population activity from the cross-validation set onto the discriminant axes. This procedure was repeated for 100 iterations, with each iteration using a different random split of the data for fitting and cross-validation.

For TDR, condition-averaged population responses across all stimulus conditions and time points were projected onto orthogonalized CDs (QR decomposition). Variance along each CD was computed across condition $\times$ time samples. The fraction of variance explained by each CD was calculated as the variance along that CD divided by the sum of variances across all CDs within the TDR subspace. Reported values reflect the means across cross-validation iterations; error bars represent SEMs across iterations.

Manifold metrics

From PCA (Fig. 3), for each cross-validation iteration and time point, we used the projected stimulus-condition centroids in the first three PCs (eight conditions: two directions $\times$ four coherences). Spread was defined as the square root of the summed variance across dimensions (i.e., the trace of the covariance matrix of stimulus centroids). Arc length was defined as the sum of Euclidean distances between adjacent centroids after ordering conditions by signed coherence (-1 , -0.64 , -0.32 , -0.16 , 0.16 , 0.32 , 0.64 , 1), capturing the extent of monotonic organization of graded evidence along the manifold.

For TDR (Fig. 4), manifold metrics were computed in two dimensions using the Direction_{CD} and Coherence_{CD} axes, for each cross-validation iteration and time point. Spread and arc length were computed analogously to PCA, but in the analogous two-dimensional (2D) CD space. Category distance was defined as the Euclidean distance between the mean position of the two stimulus direction categories (StimR versus StimL), computed as the distance between direction-specific means (averaged across coherences) in the (Direction_{CD}, Coherence_{CD}) joint subspace. We report an averaged metric for the last 1000 ms of the stimulus period, corresponding to the late stimulus-evoked epoch when LP-ACC population geometry has stabilized, as shown in the time-resolved analyses. For the metrics stratified by $|\Delta\text{Dir}|$ bins, we also performed linear regression for the mean metric over the same window as a function of $|\Delta\text{Dir}|$, and R^2 value was obtained for every cross-validated iteration and averaged.

Behavioral modeling

To recapitulate $|\Delta\text{Dir}|$ -scaled and lateralized optogenetic effects, we implemented a simplified two-hemisphere LP-ACC competition model that operates on the empirical trial sequence and simulates choices by Monte Carlo sampling (n_{iter} repetitions per condition). On each trial t , the model computes bilateral LP “comparator” signals from the trial-to-trial stimulus difference, defined as $\Delta d_t = x_t - x_{t-1}$, where x is the signed stimulus strength (direction $\times$ coherence). Depending on model variant, these comparator channels carry either signed or unsigned information, denoted d_L and d_R . A pooled magnitude signal U_t , derived from the absolute values of Δd_L and Δd_R , gates the probability of updating or switching according to a logistic function of U_t relative to a threshold. A Bernoulli switch variable is then sampled on each trial. Choices are generated from a logistic decision variable that depends on the sampled regime: a Stay decision variable and an Update decision variable, both decision variables include a bias reference term that depends on $|\Delta\text{Dir}|$ mismatch across hemispheres. Although the Update state decision variable included an additional $|\Delta\text{Dir}|$ -dependent gain on stimulus sensitivity. Unilateral optogenetic perturbations were modeled as selective transformations of the left or right comparator channel (gain, offset, or noise), whereas bilateral perturbation was applied in parallel to both channels, increasing U_t and thus the probability of switching while largely canceling in the opponent reference term used for choice. Simulations output per-trial probability of choosing right, probability of switching, and internal variables (Δd_L , Δd_R , U) for comparison to behavioral summaries. We provide more details about model construction in supplementary text, note 2.

REFERENCES AND NOTES

1. C. Summerfield, F. P. de Lange, Expectation in perceptual decision making: Neural and computational mechanisms. *Nat. Rev. Neurosci.* **15**, 745–756 (2014). doi: [10.1038/nrn3838](https://doi.org/10.1038/nrn3838); pmid: [25315388](https://pubmed.ncbi.nlm.nih.gov/25315388/)
2. D. van Moorselaar, E. Lammers, E. Cordesius, H. A. Slagter, Neural mechanisms underlying expectation-dependent inhibition of distracting information. *eLife* **9**, e61048 (2020). doi: [10.7554/eLife.61048](https://doi.org/10.7554/eLife.61048); pmid: [33320084](https://pubmed.ncbi.nlm.nih.gov/33320084/)
3. L. Chelazzi, F. Marini, D. Pascucci, M. Turatto, Getting rid of visual distractors: The why, when, how, and where. *Curr. Opin. Psychol.* **29**, 135–147 (2019). doi: [10.1016/j.copsyc.2019.02.004](https://doi.org/10.1016/j.copsyc.2019.02.004); pmid: [30856512](https://pubmed.ncbi.nlm.nih.gov/30856512/)
4. M. V. Bartsch, C. Merkel, M. A. Schoenfeld, J.-M. Hopf, Attention expedites target selection by prioritizing the neural processing of distractor features. *Commun. Biol.* **4**, 814 (2021). doi: [10.1038/s42003-021-02305-9](https://doi.org/10.1038/s42003-021-02305-9); pmid: [34188169](https://pubmed.ncbi.nlm.nih.gov/34188169/)
5. I. C. Fiebelkorn, S. Kastner, Functional specialization in the attention network. *Annu. Rev. Psychol.* **71**, 221–249 (2020). doi: [10.1146/annurev-psych-010418-103429](https://doi.org/10.1146/annurev-psych-010418-103429); pmid: [31514578](https://pubmed.ncbi.nlm.nih.gov/31514578/)
6. R. D. Rafal, M. I. Posner, Deficits in human visual spatial attention following thalamic lesions. *Proc. Natl. Acad. Sci. U.S.A.* **84**, 7349–7353 (1987). doi: [10.1073/pnas.84.20.7349](https://doi.org/10.1073/pnas.84.20.7349); pmid: [3478697](https://pubmed.ncbi.nlm.nih.gov/3478697/)
7. I. C. Fiebelkorn, S. Kastner, The puzzling pulvinar. *Neuron* **101**, 201–203 (2019). doi: [10.1016/j.neuron.2018.12.032](https://doi.org/10.1016/j.neuron.2018.12.032); pmid: [30653933](https://pubmed.ncbi.nlm.nih.gov/30653933/)
8. S. Kastner, M. A. Pinsk, Visual attention as a multilevel selection process. *Cogn. Affect. Behav. Neurosci.* **4**, 483–500 (2004). doi: [10.3758/CABN.4.4.483](https://doi.org/10.3758/CABN.4.4.483); pmid: [15849892](https://pubmed.ncbi.nlm.nih.gov/15849892/)
9. J. C. Snow, H. A. Allen, R. D. Rafal, G. W. Humphreys, Impaired attentional selection following lesions to human pulvinar: Evidence for homology between human and monkey. *Proc. Natl. Acad. Sci. U.S.A.* **106**, 4054–4059 (2009). doi: [10.1073/pnas.0810086106](https://doi.org/10.1073/pnas.0810086106); pmid: [19237580](https://pubmed.ncbi.nlm.nih.gov/19237580/)
10. C. Casanova, L. M. Chalupa, The dorsal lateral geniculate nucleus and the pulvinar as essential partners for visual cortical functions. *Front. Neurosci.* **17**, 1258393 (2023). doi: [10.3389/fnins.2023.1258393](https://doi.org/10.3389/fnins.2023.1258393); pmid: [37712093](https://pubmed.ncbi.nlm.nih.gov/37712093/)
11. N. Cortes, H. J. Ladret, R. Abbas-Farishta, C. Casanova, The pulvinar as a hub of visual processing and cortical integration. *Trends Neurosci.* **47**, 120–134 (2024). doi: [10.1016/j.tins.2023.11.008](https://doi.org/10.1016/j.tins.2023.11.008); pmid: [38143202](https://pubmed.ncbi.nlm.nih.gov/38143202/)
12. R. C. O'Reilly, J. L. Russin, M. Zolfaghar, J. Rohrlach, Deep Predictive Learning in Neocortex and Pulvinar. *J. Cogn. Neurosci.* **33**, 1158–1196 (2021). doi: [10.1162/jocn_a_01708](https://doi.org/10.1162/jocn_a_01708); pmid: [34428793](https://pubmed.ncbi.nlm.nih.gov/34428793/)
13. R. Kanai, Y. Komura, S. Shipp, K. Friston, Cerebral hierarchies: Predictive processing, precision and the pulvinar. *Phil. Trans. R. Soc. B* **370**, 20140169 (2015). doi: [10.1098/rstb.2014.0169](https://doi.org/10.1098/rstb.2014.0169); pmid: [25823866](https://pubmed.ncbi.nlm.nih.gov/25823866/)
14. I. C. Fiebelkorn, M. A. Pinsk, S. Kastner, A dynamic interplay within the frontoparietal network underlies rhythmic spatial attention. *Neuron* **99**, 842–853.e8 (2018). doi: [10.1016/j.neuron.2018.07.038](https://doi.org/10.1016/j.neuron.2018.07.038); pmid: [30138590](https://pubmed.ncbi.nlm.nih.gov/30138590/)
15. S. Kastner, I. C. Fiebelkorn, M. K. Eradath, Dynamic pulvino-cortical interactions in the primate attention network. *Curr. Opin. Neurobiol.* **65**, 10–19 (2020). doi: [10.1016/j.conb.2020.08.002](https://doi.org/10.1016/j.conb.2020.08.002); pmid: [32942125](https://pubmed.ncbi.nlm.nih.gov/32942125/)
16. Y. B. Saalmann, M. A. Pinsk, L. Wang, X. Li, S. Kastner, The pulvinar regulates information transmission between cortical areas based on attention demands. *Science* **337**, 753–756 (2012). doi: [10.1126/science.1223082](https://doi.org/10.1126/science.1223082); pmid: [22879517](https://pubmed.ncbi.nlm.nih.gov/22879517/)
17. I. C. Fiebelkorn, M. A. Pinsk, S. Kastner, The mediadorsal pulvinar coordinates the macaque fronto-parietal network during rhythmic spatial attention. *Nat. Commun.* **10**, 215 (2019). doi: [10.1038/s41467-018-08151-4](https://doi.org/10.1038/s41467-018-08151-4); pmid: [30644391](https://pubmed.ncbi.nlm.nih.gov/30644391/)
18. J. Fischer, D. Whitney, Attention gates visual coding in the human pulvinar. *Nat. Commun.* **3**, 1051 (2012). doi: [10.1038/ncomms2054](https://doi.org/10.1038/ncomms2054); pmid: [22968697](https://pubmed.ncbi.nlm.nih.gov/22968697/)
19. H. Zhou, R. J. Schafer, R. Desimone, Pulvinar-cortex interactions in vision and attention. *Neuron* **89**, 209–220 (2016). doi: [10.1016/j.neuron.2015.11.034](https://doi.org/10.1016/j.neuron.2015.11.034); pmid: [26748092](https://pubmed.ncbi.nlm.nih.gov/26748092/)
20. B. O. F. de Souza, N. Cortes, C. Casanova, Pulvinar Modulates Contrast Responses in the Visual Cortex as a Function of Cortical Hierarchy. *Cereb. Cortex* **30**, 1068–1086 (2020). doi: [10.1093/cercor/bhz149](https://doi.org/10.1093/cercor/bhz149); pmid: [31408095](https://pubmed.ncbi.nlm.nih.gov/31408095/)
21. N. Cortes, C. van Vreeswijk, The role of pulvinar in the transmission of information in the visual hierarchy. *Front. Comput. Neurosci.* **6**, 29 (2012). doi: [10.3389/fncom.2012.00029](https://doi.org/10.3389/fncom.2012.00029); pmid: [22654750](https://pubmed.ncbi.nlm.nih.gov/22654750/)
22. B. A. Olshausen, C. H. Anderson, D. C. Van Essen, A neurobiological model of visual attention and invariant pattern recognition based on dynamic routing of information. *J. Neurosci.* **13**, 4700–4719 (1993). doi: [10.1523/JNEUROSCI.13-11-04700.1993](https://doi.org/10.1523/JNEUROSCI.13-11-04700.1993); pmid: [8229193](https://pubmed.ncbi.nlm.nih.gov/8229193/)
23. C. Bennett et al., Higher-order thalamic circuits channel parallel streams of visual information in mice. *Neuron* **102**, 477–492.e5 (2019). doi: [10.1016/j.neuron.2019.02.010](https://doi.org/10.1016/j.neuron.2019.02.010); pmid: [30850257](https://pubmed.ncbi.nlm.nih.gov/30850257/)
24. Y. N. Leow et al., Brain-wide mapping of inputs to the mouse lateral posterior (LP/Pulvinar) thalamus-anterior cingulate cortex network. *J. Comp. Neurol.* **530**, 1992–2013 (2022). doi: [10.1002/cne.25317](https://doi.org/10.1002/cne.25317); pmid: [35383929](https://pubmed.ncbi.nlm.nih.gov/35383929/)
25. A. Blot et al., Visual intracortical and transthalamic pathways carry distinct information to cortical areas. *Neuron* **109**, 1996–2008.e6 (2021). doi: [10.1016/j.neuron.2021.04.017](https://doi.org/10.1016/j.neuron.2021.04.017); pmid: [33979633](https://pubmed.ncbi.nlm.nih.gov/33979633/)
26. N. P. Bichot, J. D. Schall, Effects of similarity and history on neural mechanisms of visual selection. *Nat. Neurosci.* **2**, 549–554 (1999). doi: [10.1038/9205](https://doi.org/10.1038/9205); pmid: [10448220](https://pubmed.ncbi.nlm.nih.gov/10448220/)
27. M. Sarafyazd, M. Jazayeri, Hierarchical reasoning by neural circuits in the frontal cortex. *Science* **364**, eaav8911 (2019). doi: [10.1126/science.aav8911](https://doi.org/10.1126/science.aav8911); pmid: [31097640](https://pubmed.ncbi.nlm.nih.gov/31097640/)
28. J. M. Hyman, C. B. Holroyd, J. K. Seamans, A novel neural prediction error found in anterior cingulate cortex ensembles. *Neuron* **95**, 447–456.e3 (2017). doi: [10.1016/j.neuron.2017.06.021](https://doi.org/10.1016/j.neuron.2017.06.021); pmid: [28689983](https://pubmed.ncbi.nlm.nih.gov/28689983/)
29. A. Stolyarova et al., Contributions of anterior cingulate cortex and basolateral amygdala to decision confidence and learning under uncertainty. *Nat. Commun.* **10**, 4704 (2019). doi: [10.1038/s41467-019-12725-1](https://doi.org/10.1038/s41467-019-12725-1); pmid: [31624264](https://pubmed.ncbi.nlm.nih.gov/31624264/)
30. N. Cole, M. Harvey, D. Myers-Joseph, A. Gilra, A. G. Khan, Prediction-error signals in anterior cingulate cortex drive task-switching. *Nat. Commun.* **15**, 7088 (2024). doi: [10.1038/s41467-024-51368-9](https://doi.org/10.1038/s41467-024-51368-9); pmid: [39154045](https://pubmed.ncbi.nlm.nih.gov/39154045/)
31. B. Y. Hayden, S. R. Heilbronner, J. M. Pearson, M. L. Platt, Surprise signals in anterior cingulate cortex: Neuronal encoding of unsigned reward prediction errors driving

- adjustment in behavior. *J. Neurosci.* **31**, 4178–4187 (2011). doi: [10.1523/JNEUROSCI.4652-10.2011](https://doi.org/10.1523/JNEUROSCI.4652-10.2011); pmid: [21411658](https://pubmed.ncbi.nlm.nih.gov/21411658/)
32. M. Economides, M. Guitart-Masip, Z. Kurth-Nelson, R. J. Dolan, Anterior cingulate cortex instigates adaptive switches in choice by integrating immediate and delayed components of value in ventromedial prefrontal cortex. *J. Neurosci.* **34**, 3340–3349 (2014). doi: [10.1523/JNEUROSCI.4313-13.2014](https://doi.org/10.1523/JNEUROSCI.4313-13.2014); pmid: [24573291](https://pubmed.ncbi.nlm.nih.gov/24573291/)
 33. M. P. Karlsson, D. G. R. Tervo, A. Y. Karpova, Network resets in medial prefrontal cortex mark the onset of behavioral uncertainty. *Science* **338**, 135–139 (2012). doi: [10.1126/science.1226518](https://doi.org/10.1126/science.1226518); pmid: [23042898](https://pubmed.ncbi.nlm.nih.gov/23042898/)
 34. N. Kolling *et al.*, Value, search, persistence and model updating in anterior cingulate cortex. *Nat. Neurosci.* **19**, 1280–1285 (2016). doi: [10.1038/nn.4382](https://doi.org/10.1038/nn.4382); pmid: [27669988](https://pubmed.ncbi.nlm.nih.gov/27669988/)
 35. J. Homman-Ludye, J. A. Bourne, The medial pulvinar: Function, origin and association with neurodevelopmental disorders. *J. Anat.* **235**, 507–520 (2019). doi: [10.1111/joa.12932](https://doi.org/10.1111/joa.12932); pmid: [30657169](https://pubmed.ncbi.nlm.nih.gov/30657169/)
 36. S. Furutachi, A. D. Franklin, A. M. Aldea, T. D. Mrsic-Flogel, S. B. Hofer, Cooperative thalamocortical circuit mechanism for sensory prediction errors. *Nature* **633**, 398–406 (2024). doi: [10.1038/s41586-024-07851-w](https://doi.org/10.1038/s41586-024-07851-w); pmid: [39198646](https://pubmed.ncbi.nlm.nih.gov/39198646/)
 37. S. K. Miura, M. Scanziani, Distinguishing externally from saccade-induced motion in visual cortex. *Nature* **610**, 135–142 (2022). doi: [10.1038/s41586-022-05196-w](https://doi.org/10.1038/s41586-022-05196-w); pmid: [36104560](https://pubmed.ncbi.nlm.nih.gov/36104560/)
 38. M. M. Roth *et al.*, Thalamic nuclei convey diverse contextual information to layer 1 of visual cortex. *Nat. Neurosci.* **19**, 299–307 (2016). doi: [10.1038/nn.4197](https://doi.org/10.1038/nn.4197); pmid: [26691828](https://pubmed.ncbi.nlm.nih.gov/26691828/)
 39. F. Hu *et al.*, Prefrontal corticotectal neurons enhance visual processing through the superior colliculus and pulvinar thalamus. *Neuron* **104**, 1141–1152.e4 (2019). doi: [10.1016/j.neuron.2019.09.019](https://doi.org/10.1016/j.neuron.2019.09.019); pmid: [31668485](https://pubmed.ncbi.nlm.nih.gov/31668485/)
 40. Q. Fang *et al.*, A differential circuit via retino-colliculo-pulvinar pathway enhances feature selectivity in visual cortex through surround suppression. *Neuron* **105**, 355–369.e6 (2020). doi: [10.1016/j.neuron.2019.10.027](https://doi.org/10.1016/j.neuron.2019.10.027); pmid: [31812514](https://pubmed.ncbi.nlm.nih.gov/31812514/)
 41. D. Gupta, B. DePasquale, C. D. Kopec, C. D. Brody, Trial-history biases in evidence accumulation can give rise to apparent lapses in decision-making. *Nat. Commun.* **15**, 662 (2024). doi: [10.1038/s41467-024-44880-5](https://doi.org/10.1038/s41467-024-44880-5); pmid: [38253526](https://pubmed.ncbi.nlm.nih.gov/38253526/)
 42. C. Findling *et al.*, Brain-wide representations of prior information in mouse decision-making. *Nature* **645**, 192–200 (2025). doi: [10.1038/s41586-025-09226-1](https://doi.org/10.1038/s41586-025-09226-1); pmid: [40903597](https://pubmed.ncbi.nlm.nih.gov/40903597/)
 43. E. J. Hwang, J. E. Dahlen, M. Mukundan, T. Komiyama, History-based action selection bias in posterior parietal cortex. *Nat. Commun.* **8**, 1242 (2017). doi: [10.1038/s41467-017-01356-z](https://doi.org/10.1038/s41467-017-01356-z); pmid: [29089500](https://pubmed.ncbi.nlm.nih.gov/29089500/)
 44. A. Akrami, C. D. Kopec, M. E. Diamond, C. D. Brody, Posterior parietal cortex represents sensory history and mediates its effects on behaviour. *Nature* **554**, 368–372 (2018). doi: [10.1038/nature25510](https://doi.org/10.1038/nature25510); pmid: [29414944](https://pubmed.ncbi.nlm.nih.gov/29414944/)
 45. G. Okazawa, C. E. Hatch, A. Mancoo, C. K. Machens, R. Kiani, Representational geometry of perceptual decisions in the monkey parietal cortex. *Cell* **184**, 3748–3761.e18 (2021). doi: [10.1016/j.cell.2021.05.022](https://doi.org/10.1016/j.cell.2021.05.022); pmid: [34171308](https://pubmed.ncbi.nlm.nih.gov/34171308/)
 46. G. Okazawa, R. Kiani, Neural mechanisms that make perceptual decisions flexible. *Annu. Rev. Physiol.* **85**, 191–215 (2023). doi: [10.1146/annurev-physiol-031722-024731](https://doi.org/10.1146/annurev-physiol-031722-024731); pmid: [36343603](https://pubmed.ncbi.nlm.nih.gov/36343603/)
 47. N. Kriegeskorte, X.-X. Wei, Neural tuning and representational geometry. *Nat. Rev. Neurosci.* **22**, 703–718 (2021). doi: [10.1038/s41583-021-00502-3](https://doi.org/10.1038/s41583-021-00502-3); pmid: [34522043](https://pubmed.ncbi.nlm.nih.gov/34522043/)
 48. Y. Wang *et al.*, A frontal transcallosal inhibition loop mediates interhemispheric balance in visuospatial processing. *Nat. Commun.* **14**, 5213 (2023). doi: [10.1038/s41467-023-40985-5](https://doi.org/10.1038/s41467-023-40985-5); pmid: [37626171](https://pubmed.ncbi.nlm.nih.gov/37626171/)
 49. J. C. Erlich, B. W. Brunton, C. A. Duan, T. D. Hanks, C. D. Brody, Distinct effects of prefrontal and parietal cortex inactivations on an accumulation of evidence task in the rat. *eLife* **4**, e05457 (2015). doi: [10.7554/eLife.05457](https://doi.org/10.7554/eLife.05457); pmid: [25869470](https://pubmed.ncbi.nlm.nih.gov/25869470/)
 50. T. D. Hanks *et al.*, Distinct relationships of parietal and prefrontal cortices to evidence accumulation. *Nature* **520**, 220–223 (2015). doi: [10.1038/nature14066](https://doi.org/10.1038/nature14066); pmid: [25600270](https://pubmed.ncbi.nlm.nih.gov/25600270/)
 51. J.-H. Kim, D.-H. Ma, E. Jung, I. Choi, S.-H. Lee, Gated feedforward inhibition in the frontal cortex releases goal-directed action. *Nat. Neurosci.* **24**, 1452–1464 (2021). doi: [10.1038/s41593-021-00910-9](https://doi.org/10.1038/s41593-021-00910-9); pmid: [34413512](https://pubmed.ncbi.nlm.nih.gov/34413512/)
 52. S. J. Cruikshank *et al.*, Thalamic control of layer 1 circuits in prefrontal cortex. *J. Neurosci.* **32**, 17813–17823 (2012). doi: [10.1523/JNEUROSCI.3231-12.2012](https://doi.org/10.1523/JNEUROSCI.3231-12.2012); pmid: [23223300](https://pubmed.ncbi.nlm.nih.gov/23223300/)
 53. A. J. Lee *et al.*, Canonical Organization of Layer 1 Neuron-Led Cortical Inhibitory and Disinhibitory Interneuronal Circuits. *Cereb. Cortex* **25**, 2114–2126 (2015). doi: [10.1093/cercor/bhu020](https://doi.org/10.1093/cercor/bhu020); pmid: [24554728](https://pubmed.ncbi.nlm.nih.gov/24554728/)
 54. B. Schuman *et al.*, Four unique interneuron populations reside in neocortical layer 1. *J. Neurosci.* **39**, 125–139 (2019). doi: [10.1523/JNEUROSCI.1613-18.2018](https://doi.org/10.1523/JNEUROSCI.1613-18.2018); pmid: [30413647](https://pubmed.ncbi.nlm.nih.gov/30413647/)
 55. L. Z. Fan *et al.*, All-optical electrophysiology reveals the role of lateral inhibition in sensory processing in cortical layer 1. *Cell* **180**, 521–535.e18 (2020). doi: [10.1016/j.cell.2020.01.001](https://doi.org/10.1016/j.cell.2020.01.001); pmid: [31978320](https://pubmed.ncbi.nlm.nih.gov/31978320/)
 56. G. Ma *et al.*, Hierarchy in sensory processing reflected by innervation balance on cortical interneurons. *Sci. Adv.* **7**, eabf5676 (2021). doi: [10.1126/sciadv.abf5676](https://doi.org/10.1126/sciadv.abf5676); pmid: [33990327](https://pubmed.ncbi.nlm.nih.gov/33990327/)
 57. Y. B. Saalman, S. Kastner, The cognitive thalamus. *Front. Syst. Neurosci.* **9**, 39 (2015). doi: [10.3389/fnsys.2015.00039](https://doi.org/10.3389/fnsys.2015.00039); pmid: [25852498](https://pubmed.ncbi.nlm.nih.gov/25852498/)
 58. J. Jaramillo, J. F. Mejias, X.-J. Wang, Engagement of pulvino-cortical feedforward and feedback pathways in cognitive computations. *Neuron* **101**, 321–336.e9 (2019). doi: [10.1016/j.neuron.2018.11.023](https://doi.org/10.1016/j.neuron.2018.11.023); pmid: [30553546](https://pubmed.ncbi.nlm.nih.gov/30553546/)
 59. Y. Komura, A. Nikkuni, N. Hirashima, T. Uetake, A. Miyamoto, Responses of pulvinar neurons reflect a subject's confidence in visual categorization. *Nat. Neurosci.* **16**, 749–755 (2013). doi: [10.1038/nn.3393](https://doi.org/10.1038/nn.3393); pmid: [23666179](https://pubmed.ncbi.nlm.nih.gov/23666179/)
 60. J. Friedrich, P. Zhou, L. Paninski, Fast online deconvolution of calcium imaging data. *PLOS Comput. Biol.* **13**, e1005423 (2017). doi: [10.1371/journal.pcbi.1005423](https://doi.org/10.1371/journal.pcbi.1005423); pmid: [28291787](https://pubmed.ncbi.nlm.nih.gov/28291787/)
 61. A. Giovannucci *et al.*, CalmAn an open source tool for scalable calcium imaging data analysis. *eLife* **8**, e38173 (2019). doi: [10.7554/eLife.38173](https://doi.org/10.7554/eLife.38173); pmid: [30652683](https://pubmed.ncbi.nlm.nih.gov/30652683/)
 62. V. Mante, D. Sussillo, K. V. Shenoy, W. T. Newsome, Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature* **503**, 78–84 (2013). doi: [10.1038/nature12742](https://doi.org/10.1038/nature12742); pmid: [24201281](https://pubmed.ncbi.nlm.nih.gov/24201281/)
 63. Y. N. Leow *et al.*, Code and Source Data for 'A thalamo-cortical circuit for contrastive coding of sensory history'. Zenodo (2026). doi: [10.5281/zenodo.20267242](https://doi.org/10.5281/zenodo.20267242)

ACKNOWLEDGMENTS

We thank T. Johns for laboratory management and members of the Sur laboratory for many discussions and comments. We also thank L. Petreanu and former laboratory members for pointers on beginning the 2AFC RDK discrimination behavior. **Funding:** This study was supported by National Institutes of Health (NIH) grants R01MH126351, R01MH133066, and R01NS130361 to M.S.; Multidisciplinary University Research Initiatives grant W911NF2110328 to M.S.; the Picower Institute Innovation Fund to M.S.; the Simons Foundation Autism Research Initiative through the Simons Center for the Social Brain to M.S.; and an A*STAR National Science Scholarship (BS-PhD) to Y.N.L. **Author contributions:** Conceptualization: Y.N.L., M.S.; Funding acquisition: M.S.; Investigation: Y.N.L., A.N., A.B., T.L., S.A.-R., M.J., M.S.; Supervision: M.S., M.J.; Visualization: Y.N.L.; Writing – original draft: Y.N.L.; Writing – review & editing: Y.N.L., S.A.-R., A.N., M.S., M.J. **Competing interests:** The authors declare that they have no competing interests. **Data, code, and materials availability:** All data necessary for evaluating the conclusions are available in the main text or the supplementary materials. Code used for analyses and source data are deposited and available through Zenodo (63). **License information:** Copyright © 2026 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. <https://www.science.org/about/science-licenses-journal-article-reuse>

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.aeg4720

Supplementary Text; Figs. S1 to S17; Tables S1 to S5; MDAR Reproducibility Checklist

Submitted 14 February 2026; accepted 26 May 2026

10.1126/science.aeg4720



A thalamocortical circuit encoding deviations from sensory history

Yi Ning Leow, Arundhati Natesan, Alexandria Barlowe, Sofie Åhrlund-Richter, Tianyu (Cindy) Luo, Mehrdad Jazayeri, and Mriganka Sur

Science **393** (6812), eaeg4720. DOI: 10.1126/science.aeg4720

Editor's summary

Past experiences guide future decisions. Leow *et al.* studied the brain circuits mediating experience-dependent decision-making in mice. Using a combination of behavioral testing, optogenetic manipulation, calcium imaging, and computational techniques, the study reports that the thalamic lateral posterior nucleus-to-anterior cingulate cortex (LP-ACC) pathway uses information from the previous trial to guide adaptive decision-making. Manipulating LP-ACC disrupted the ability to use prior experiences to guide decisions. Further, calcium-imaging experiments showed that LP-ACC axons represent stimuli in a low-dimensional space, with each stimulus expressed as a point on a curved manifold encoding both stimulus direction and difficulty. This study highlights the key role of the thalamus in high-level cognition. —Mattia Maroso

View the article online

<https://www.science.org/doi/10.1126/science.aeg4720>

Permissions

<https://www.science.org/help/reprints-and-permissions>

Use of this article is subject to the [Terms of service](#)

Science (ISSN 1095-9203) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005. The title *Science* is a registered trademark of AAAS.

Copyright © 2026 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works